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Breaking the Tabular Ceiling: Heterogeneous Graph-Aware Multimodal Fusion for M&A Synergy Prediction

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Abstract

A large body of research finds that a majority of mergers and acquisitions fail to create value for shareholders, yet the tools used to predict these outcomes have barely changed in decades. Most models still treat companies as isolated islands of financial data, ignoring the complex networks of suppliers and customers that actually drive their success. This dissertation builds a new kind of "graph-aware" model that sees the economy as a web of relationships. By combining traditional financial ratios with supply-chain maps and a section-by-section analysis of corporate filings, I show that we can identify value-creating deals more accurately than before.

Using a dataset of 2,864 US acquisitions from 2000 to 2023, I built a multimodal system called HeteroGraphSAGE. The results prove that knowing a company's position in the supply chain provides a "topological alpha"—a predictive edge that financial numbers alone can't capture. I also found that natural language processing (NLP) is a double-edged sword: if you just "dump" all the text from a company report into a model, accuracy actually drops. However, when you specifically compare strategic plans (MD&A) against risk disclosures, the model begins to understand the true trade-offs of a deal. While predicting the exact dollar value of a merger remains incredibly difficult, this study demonstrates that we can reliably predict the **direction** of the outcome, providing a clearer way to triage deals in a noisy market.

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Finally, thank you to my family and friends. To those who listened to my frustrated rants about "negative R-squared" values and the "M2 reversal" over coffee, and to those who simply reminded me to take a walk and step away from the terminal—thank you. Your support made the long nights of data engineering a lot more manageable.

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Chapter 1: Introduction

1.1 The Problem: A Century of Predictable Failure

Every year, hundreds of companies announce mergers and acquisitions worth trillions of dollars. Each deal rests on the same fundamental bet: that the combined entity will generate more economic value than the two firms could produce independently. The gap between that bet and its typical outcome is one of the most durable findings in modern corporate finance. A large body of evidence suggests that most acquisitions fail to generate value for acquirer shareholders, a failure rate that has remained statistically stable across a century of takeover activity and through every wave of increasingly sophisticated due diligence technology (Martynova and Renneboog, 2008). The question this dissertation asks is not **why** individual deals fail - the behavioural and market-timing explanations for that are well-established. The question is structural: what information do all existing predictive frameworks systematically miss, and is that information recoverable?

The answer proposed here is that existing models fail not because of insufficient computational power, but because of a flawed representational assumption. Every major class of quantitative M&A predictor - from logistic regression on financial ratios to gradient-boosted trees and transformer-based NLP pipelines - treats each firm as an isolated data point. This means that no matter how sophisticated the algorithm, it operates in a feature space that is topologically flat: it sees the numbers on the balance sheet but cannot see the economic ecosystem the firm inhabits. A firm's largest customer may be on the verge of bankruptcy. Its target may share fragile single-source suppliers. The combined entity may create a bottleneck in a critical industrial network. None of these facts appear in a financial ratio vector, and none can be recovered by adding more layers to the same architecture.

1.2 Aims and Objectives

The central aim of this study is to construct, evaluate, and interpret a heterogeneous multimodal machine learning framework for M&A synergy prediction that transcends traditional tabular limitations by integrating financial, textual, and topological data.

The project was guided by the following core objectives:

1. **Literature Synthesis:** To conduct a comprehensive review of existing econometric, NLP, and network-based M&A prediction models, identifying the structural limitations of isolated analytical streams.
2. **Multimodal Dataset Construction:** To engineer a unified dataset combining historical M&A financial fundamentals, section-specific 10-K regulatory filings, and inter-firm supply-chain relationship graphs.
3. **Architectural Development:** To design and train a late-fusion neural architecture capable of processing this heterogeneous feature space.

4. **Benchmark Evaluation:** To rigorously evaluate the multimodal model's predictive performance (AUC-ROC) against a standard financial-only baseline to quantify the incremental value of non-financial signals.
5. **Interpretability and Feature Attribution:** To utilise SHAP (SHapley Additive exPlanations) to deconstruct model predictions, identifying the precise economic drivers of synergy and interpreting the interaction between financial fundamentals and ecosystem topology.

1.3 The Proposed Solution

This dissertation proposes and empirically tests a multimodal late-fusion architecture designed to break through the tabular ceiling by encoding three complementary information modalities simultaneously:

- **Block A - Financial Fundamentals:** 56 ratio-level features drawn from Yahoo Finance, covering acquirer and target leverage, liquidity, profitability, and deal structure. This is the baseline information available to every prior quantitative model.
- **Block B - Section-Aware Textual Semantics:** FinBERT-encoded embeddings from the Management Discussion & Analysis (MD&A) and Risk Factors sections of pre-announcement 10-K filings, processed separately so that their opposing economic signals are not cancelled by aggregation.
- **Block C - Heterogeneous Graph Topology:** Node embeddings derived from a two-hop HeteroGraphSAGE model trained on Bloomberg SPLC supply-chain relationship data, encoding each firm's structural position within the industrial network it actually inhabits.

The three blocks are fused via late concatenation and evaluated across a dual-pipeline framework: a classifier pipeline that predicts the **direction** of acquirer Cumulative Abnormal Return (CAR) at announcement, and a regression pipeline that attempts to predict CAR **magnitude**. As Chapter 4 demonstrates, these two pipelines resolve differently - and that difference is itself a substantive finding. To ensure absolute methodological rigour, the pipeline enforces strict temporal splitting and an 11-day event-window embargo to explicitly prevent target leakage and market microstructure contamination.

1.4 Research Hypotheses

The study tests three formal hypotheses derived from the four knowledge streams surveyed in Chapter 2. Each hypothesis corresponds to one of the three modality blocks and is designed to be falsifiable within the available dataset.

H1 - The Topological Alpha Hypothesis: The inclusion of heterogeneous supply-chain graph topology will yield a statistically significant improvement in directional deal discrimination (AUC-ROC) over a financial-only baseline, proving that topological data yields orthogonal predictive alpha.

H2 - The Semantic Divergence Hypothesis: The relationship between textual similarity and announcement returns is conditional on document section. MD&A similarity (strategic fit) correlates positively with CAR; Risk Factor similarity (shared liability exposure) correlates negatively. These opposing effects are mathematically suppressed when the two sections are pooled.

H3 - The Topological Arbitrage Hypothesis: Acquirers with high betweenness centrality in the supply-chain graph exhibit statistically compressed variance in announcement return outcomes relative to peripheral acquirers, as measured by Levene's test across centrality quantile groups.

1.5 What This Study Contributes

The academic contribution of this work lies at the intersection of three fields that have not previously been unified in the context of post-merger synergy prediction. Supply chain finance, financial NLP, and heterogeneous graph neural networks have each advanced substantially as independent research programmes. This dissertation is, to the best of current knowledge, the first study to direct their combination at binary CAR direction classification as a proxy for M&A synergy.

The contribution is not only architectural. By reporting the M2 reversal - the empirical finding that adding aggregated FinBERT text **degrades** prediction quality - the study provides a methodological warning to future researchers that NLP is not automatically useful in M&A prediction unless section semantics are preserved. By reporting negative R^2 values on the regression pipeline, the study defines the boundary between what multimodal architecture can and cannot achieve in this domain. And by building the Deal Intelligence Terminal, the study operationalises the theoretical framework into a live, reproducible research artefact that renders all empirical claims interactively.

1.6 Structure of the Dissertation

The dissertation proceeds as follows. Chapter 2 builds a sustained critical argument across four interconnected knowledge streams, demonstrating that each wave of prior scholarship produced an asymptotic ceiling and identifying precisely the structural information each paradigm discarded. Chapter 3 details the full research design, data pipeline, feature construction, and hypothesis-testing protocol. Chapter 4 reports the empirical findings across both pipelines, including the negative results that establish the boundary conditions of the architecture. Chapter 5 evaluates both the research product and the research process honestly, contextualising performance against the literature and identifying what should be done differently in future work. Chapter 6 synthesises the findings and places the contribution in the context of the broader financial machine learning research agenda.

Chapter 2: Literature Review

2.1 The Central Argument: A Gap Three Decades Wide

Every year, thousands of companies announce mergers and acquisitions - deals in which one firm pays a premium to absorb another, betting that the combined entity will be worth more than the sum of its parts. Yet decades of evidence suggest that this bet fails more often than it succeeds. The central question motivating this study is simple: if this failure rate has persisted across a century of increasingly sophisticated corporate due diligence, what information are existing analytical frameworks systematically missing?

This review builds a sustained argument across four interconnected knowledge streams, each representing a distinct wave of scholarship that has tried - and partially failed - to answer that question. Rather than a neutral survey, each stream is examined for what it structurally could not achieve, with each identified gap mapped directly to one of the three hypotheses tested in this dissertation.

1. **Stream I - The M&A Paradox and the Failure of Valuation Theory:** diagnoses why synergy prediction is genuinely difficult and establishes that the root cause is not a lack of computing power, but a structural information blindspot.
2. **Stream II - The Tabular Paradigm:** traces the full history of quantitative M&A models - from logistic regression through gradient-boosted trees and early deep neural networks - and demonstrates that every generation inherited the same core architectural limitation, producing an asymptotic accuracy ceiling no amount of algorithmic sophistication could break through.
3. **Stream III - The Semantic Turn:** examines how natural language processing entered the M&A prediction landscape and why, despite genuinely powerful tools, the field systematically pointed them at the wrong question - and even when pointed correctly, fell into a structural trap that cancelled the very richness it sought to exploit.
4. **Stream IV - The Topological Turn:** engages with supply chain finance, corporate network theory, and graph neural network research to establish that the structure of a firm's external relationships - its suppliers, customers, and competitors - carries economic information about deal value that is, by mathematical necessity, invisible to any model that treats each firm as an isolated data point.

The review concludes that these four streams converge on a single architectural response: a heterogeneous graph model that simultaneously fuses financial fundamentals (Block A), section-conditioned textual embeddings (Block B), and supply-chain and competition graph topology (Block C). As of 2025, no published study has directed this combined architecture at post-merger synergy outcome prediction.

2.2 Stream I: The M&A Paradox and the Failure of Valuation Theory

2.2.1 The Empirical Record: Why Deals Keep Failing

Mergers and acquisitions are the primary mechanism through which capital is reallocated across the global economy. When a company acquires another, the underlying premise is that the combined entity will generate more value than the two firms could independently - through cost savings, revenue growth, market power, or access to new capabilities. This theoretical surplus is called "synergy."

The empirical record, however, tells a more troubling story. Martynova and Renneboog (2008) documented that between 70% and 90% of acquisitions fail to generate value for the acquirer's shareholders - a finding replicated across multiple independent research programmes (Christensen *et al.*, 2011). More striking than the magnitude of this failure rate is its persistence: it has remained statistically stable across entirely different economic eras, from the conglomerate wave of the 1960s through the technology boom of the late 1990s and the post-financial-crisis consolidation wave of the 2010s (Martynova and Renneboog, 2008). A failure rate that survives such varied conditions is unlikely to be cyclical; it points toward something structurally wrong with how deals are evaluated.

Bradley, Desai and Kim (1988) established a revealing asymmetry in the data: while shareholders of acquired firms (the targets) captured average announcement-day abnormal returns of approximately 30%, the shareholders of acquiring firms systematically lost value, with combined deal synergy gains averaging only 7.4% of combined pre-deal firm value. The gap between what acquiring firms paid and what the market believed the combination was worth is precisely what this study attempts to predict.

2.2.2 Behavioural Distortions: When Executive Confidence Contaminates the Data

The first layer of explanation for persistent M&A failure is behavioural. Roll (1986) proposed the Hubris Hypothesis: acquiring managers systematically overestimate their own ability to extract value from a target, treating the market's price for that target as an underestimate rather than as a rational, information-weighted signal. This overconfidence drives them to pay premiums that the actual post-merger performance cannot justify - what Sirower (1997) termed the "Synergy Trap."

From a machine learning perspective, this creates a specific and irreducible data quality problem. If the acquisition premiums recorded in deal databases partly reflect executive overconfidence rather than genuine synergy potential, then any model trained to predict outcomes from those premiums is fitting a signal that has been systematically contaminated with human behavioural error. No amount of regularisation or hyperparameter tuning can remove bias that is baked into the target variable itself. This is one of the reasons this study uses Cumulative Abnormal Return (CAR) - the stock market's immediate revision of its estimate of firm value upon hearing the deal announcement - as the prediction target, rather than deal premia or accounting-based synergy measures. CAR represents the market's collective, forward-looking assessment of deal value and is considerably less susceptible to the narrative distortions of individual executives (MacKinlay, 1997).

2.2.3 Market Timing: When Deal Waves Are About Valuation, Not Strategy

A complementary source of noise in M&A data was identified by Shleifer and Vishny (2003). Their stock-market-driven acquisitions model demonstrates that companies with temporarily overvalued equity have a rational incentive to use their inflated shares as acquisition currency - essentially exchanging overpriced paper for real assets. Under this model, a substantial fraction of any M&A wave is driven by relative misvaluation rather than genuine operational complementarity between the two firms. The observed announcement-period returns, in these cases, reflect markets correcting for overvaluation rather than pricing synergy realisation. Grossman and Stiglitz (1980) established the theoretical boundary here: in a world where gathering and processing information is costly, markets cannot be perfectly efficient, and persistent information asymmetries around deal announcements are a natural and stable equilibrium. This bounded-efficiency framing is what makes the prediction problem tractable in the first place - if markets perfectly and instantaneously priced all available information, there would be no learnable signal left.

2.2.4 The Epistemological Barrier: What Due Diligence Cannot See

Even setting aside behavioural and timing distortions, the fundamental problem is that conventional due diligence is architecturally blind to certain categories of risk. Angwin (2001) documented that standard due diligence operates in organisational silos: financial, legal, and operational teams independently assess their respective domains without modelling how those domains interact. This fragmentation means that multiplicative risks are systematically missed. Consider a firm with strong cash flows that happens to source 80% of its critical components from a single supplier. Each dimension passes a standard due diligence check in isolation; the conjunction - that the entire cash-flow advantage is conditional on a single supply relationship - is invisible to teams that never share information.

Akerlof (1970) provided the theoretical foundation for why mandated disclosures cannot fully resolve this problem. His “Market for Lemons” framework shows that when buyers and sellers have asymmetric information, sellers have an incentive to obscure their vulnerabilities within the noise of standard reporting formats. A 10-K filing, however detailed, is ultimately a document a company prepares about itself. What it cannot obscure - at least not easily - is the verifiable structure of its external relationships: which firms it buys from, sells to, and competes with. This network of real economic ties is precisely what Block C in this study’s architecture encodes, and it is the one source of synergy-relevant information that is most resistant to strategic misrepresentation.

Stream I conclusion: The M&A failure rate is partly predictable from structural information that existing valuation frameworks systematically exclude. The following three streams each diagnose a specific category of excluded information and the architectural limitations that kept it out.

2.3 Stream II: The Tabular Paradigm - An Asymptotic Ceiling

2.3.1 The Econometric Era: When Linearity Assumed Too Much

Before examining the historical progression of quantitative M&A models, the structural limitation of the "tabular" paradigm must be defined. A tabular model takes a spreadsheet-style matrix of numbers - one row per deal, one column per financial ratio or deal characteristic - and learns statistical patterns that separate value-creating deals from value-destroying ones. Each deal is treated as an independent, self-contained data point. The model has no mechanism for knowing that two deals involve acquirers who share a supplier, or that the target in one deal is a customer of the acquirer in another. It sees the numbers; it does not see the relationships.

The first generation of quantitative M&A models - most influentially Palepu (1986) and Barnes (1990) - used logistic regression and multiple discriminant analysis on financial ratios. These approaches assume that each financial variable has a monotonic, linear relationship with the outcome: more of a given ratio always either increases or decreases the probability of a successful deal. Barnes (1990) demonstrated empirically that this assumption breaks down for leverage. Moderate debt signals fiscal discipline and managerial confidence; very high debt signals financial fragility. The relationship is non-monotonic - it changes direction at some threshold - and linear models fundamentally cannot represent direction changes.

More revealing still, Palepu (1986)'s own out-of-sample validation found that acquisition targets could not be predicted reliably beyond chance levels. The information contained in financial ratios was simply insufficient to reconstruct synergy outcomes — poor out-of-sample performance is the natural consequence. The consistently low pseudo-R^2 values (below 0.10) documented across econometric studies (Betton, Eckbo and Thorburn, 2008) reflect the information gap inherent to financial ratio feature spaces, not model miscalibration. This ceiling is the first empirical benchmark any successor model must demonstrably improve upon.

2.3.2 Why Machine Learning Could Not Fully Escape the Same Trap

The arrival of ensemble machine learning methods - random forests, gradient-boosted decision trees, and ultimately XGBoost - promised to resolve the non-linearity problem. Zhang *et al.* (2024) and similar studies deployed these techniques on financial ratio vectors and did report accuracy improvements over logistic baselines. Those improvements are real, but they are bounded by four structural problems that persist regardless of how sophisticated the algorithm becomes.

The Independence Ceiling: Both random forests and XGBoost operate on rows of a feature table where each row represents one deal in complete isolation. The algorithm learns patterns within individual deals but has no mechanism to propagate information between related deals. If two acquiring firms share a common supplier, and that supplier is experiencing financial distress, both deals carry a latent risk that is invisible in their individual financial ratios. Adding more features cannot resolve this — models that process each deal in isolation have no architectural mechanism for propagating information between related deals, regardless of feature richness (Hamilton, Ying and Leskovec, 2017).

Survivorship Bias: M&A datasets are almost exclusively drawn from completed transactions. Deals that were considered but abandoned - often precisely because due diligence revealed serious problems - never appear in the training data. This systematically biases estimated model coefficients toward the characteristics of deals that proceeded, rather than deals that actually created value (Betton, Eckbo and Thorburn, 2008). No machine learning algorithm can correct for data that was never collected.

Look-Ahead Contamination: Several prominent ML studies, including Zhang *et al.* (2024), construct financial features from annual filings without explicitly verifying that the fiscal year end-date precedes the deal announcement date. When a model trains on financial data that postdates the deal it is predicting, its apparent accuracy reflects information that would not have been available at the time of the decision - inflating reported performance beyond any economically realisable level.

Crystallised Analyst Bias: A common preprocessing step in these pipelines is discretising continuous financial variables into categorical bins before feeding them into tree models. This encodes whatever assumptions the analyst made when deciding where to draw the bin boundaries into the feature representation in a way that cannot be undone. The model then learns from a filtered, pre-interpreted version of the data rather than the raw signal - an automated version of the same heuristic reasoning employed by human analysts whose systematic failures are the very motivation for building quantitative models in the first place.

Taken together, these four problems create what can be described as an asymptotic accuracy ceiling: once a tabular model has captured all the linear, rank-order, and non-parametric signal available in deal-level financial features, no further improvement is achievable by making the model more sophisticated. The ceiling is a function of the information content of the feature space, not of computational power.

2.3.3 Why Deep Neural Networks Did Not Break Through

The introduction of deep multi-layer perceptrons (MLPs) and recurrent sequence models appeared to dissolve the linearity constraint entirely. Elhoseny *et al.* (2022) reported 95.8% accuracy on financial distress prediction using a deeply optimised neural network - a figure that superficially suggests the prediction problem has been solved. It has not, for a reason that is worth spelling out carefully.

Financial distress prediction and M&A synergy prediction are fundamentally different questions. The former asks: “Is this firm, on its own, likely to fail?” That question can be answered, at least in part, from a single firm’s own financial time series. The latter asks: “Will combining these two specific firms create more value than they possess separately?” That is an inherently relational question - its answer depends on the interaction between the two firms and on their positions within a broader industrial ecosystem. An MLP with ten layers and ten thousand parameters, trained on a vector of fifty financial ratios, still operates in a fifty-dimensional space that is topologically flat. It has no representation of whether the acquirer's largest customer is about to go bankrupt, or whether the target and acquirer share a fragile single-source supplier. These omissions cannot be corrected by adding more layers to the same architecture.

Stream II conclusion: Every generation of tabular model - logistic regression, random forest, XGBoost, MLP, LSTM - inherits the independence assumption from its predecessor. The asymptotic ceiling is not a function of model complexity but of the information content of the feature space. Breaking through it requires a fundamentally different kind of model: one that operates on relationships rather than rows. Stream IV addresses this. Before arriving there, Stream III examines why adding language data - an intuitively promising additional information source - proved far more complicated than the field anticipated.

2.4 Stream III: The Semantic Turn - Powerful Tools, Wrong Targets

2.4.1 The Bag-of-Words Era: A Useful Foundation with Hard Limits

Where tabular models read financial numbers, semantic models read text - in the M&A context, primarily the annual 10-K reports that US-listed firms file with the Securities and Exchange Commission. These filings contain structured, standardised sections: the Management Discussion and Analysis (MD&A), in which management describes strategy, performance, and outlook in their own words, and the Risk Factors section, in which the company enumerates specific threats and uncertainties it faces. The intuition motivating semantic M&A research is that these disclosures might reveal complementarity or conflict between acquirer and target that financial ratios alone cannot capture.

The foundational contribution to financial NLP is Loughran and McDonald (2011), who demonstrated that standard general-purpose sentiment dictionaries systematically mislabel financial language. Words like "liability," "risk," and "obligation" carry negative connotations in ordinary English but are routine, neutral legal descriptors in financial filings. Their domain-specific word lists corrected this systematic mislabelling and established a principle that remains foundational: financial text requires tools calibrated for the specific conventions of financial language. This study directly inherits that principle in its choice of FinBERT over general-purpose language models.

The bag-of-words (BoW) paradigm that dominated early financial NLP, however, carries a structural limitation that no amount of vocabulary refinement can overcome: it is compositionally blind. A BoW model represents a document as a count of how many times each word appears, with no memory of the order or context in which words appeared. The sentence "We have eliminated our exposure to risk" and the sentence "We have significant exposure to risk" produce nearly identical word-frequency vectors while conveying directly opposite strategic signals. In M&A filings, this ambiguity constitutes a systematic failure mode for documents where a single conditional clause can invert the economic meaning of a multi-billion-dollar commitment.

Despite this, a substantial body of recent research has continued to extend the BoW approach: Demers, Wang and Wu (2024) construct frequency-based lexicons for human capital disclosures; Acheampong *et al.* (2025) proxy financial constraints using lexicon-based indices; Garcia, Hu and Rohrer (2020) develop finance-specific dictionaries validated through market data. Each of these contributions improves the vocabulary but cannot address the underlying compositional blindness. Hoberg and Phillips (2016) represents the most structurally sophisticated BoW application - their text-based network industries classification used product-description cosine similarity from 10-K filings to construct time-varying firm similarity networks, demonstrating that textual proximity predicts competitive dynamics and deal likelihood. This study inherits that semantic proximity methodology directly, but redirects it from predicting whether a deal will happen to predicting whether a deal that has happened will create or destroy value, and replaces TF-IDF word vectors with contextualised FinBERT embeddings.

2.4.2 The Transformer Revolution: A Better Tool, But Often Misdirected

Devlin *et al.* (2018)'s BERT architecture marked a genuine qualitative advance in language modelling. Rather than counting word occurrences, BERT learns contextual embeddings - vector representations in which the same word receives a different numerical encoding depending on the surrounding words. This directly resolves the compositional blindness of BoW models. Araci (2019) specialized this architecture for financial language through pre-training on a large corpus of corporate filings, giving it domain-calibrated representations that a general BERT model lacks.

The challenge is not with the tools themselves but with how the field has applied them to M&A prediction. Zhao, Li and Zheng (2020) and Lohmeier and Stitz (2023) both used transformer architectures to extract semantic features from M&A-related text, but then compressed the resulting rich, high-dimensional representations into a single scalar sentiment score before feeding them into a downstream classifier. This pipeline - transformer as sophisticated word counter, followed by scalar compression - preserves almost none of the representational richness that motivated using a transformer in the first place. The architecture is, in terms of information content, not substantially different from an advanced bag-of-words approach.

More fundamentally, Hájek and Henriques (2024) and Lohmeier and Stitz (2023) both direct their NLP pipelines at predicting **deal occurrence** - whether an acquisition announcement will happen - rather than **deal outcome** - whether the announced deal will create or destroy shareholder value. These are genuinely different problems. Predicting deal occurrence requires detecting patterns in news flow that precede announcements; predicting deal outcome requires modelling the fundamental complementarity between acquirer and target across multiple dimensions. The existing NLP literature has made substantial progress on the former and has largely not addressed the latter.

2.4.3 Why Standard FinBERT Cannot Fully Recover the Textual Signal

Even correctly directed at deal outcome prediction, the standard single-document FinBERT implementation faces three structural limitations that this study directly addresses.

Frozen generic embeddings: FinBERT's pre-training objective is general financial language modelling - predicting masked words in financial text - not the specific task of discriminating strategic alignment from risk concentration. Its frozen weights cannot adapt to the semantic distinctions most relevant to synergy prediction. This study addresses this by treating the two dominant 10-K sections - MD&A and Risk Factors - as semantically distinct modalities rather than interchangeable text, exploiting a distinction FinBERT cannot make on its own.

Single-document architecture: Studies that apply FinBERT independently to the acquirer's filing or the target's filing separately miss the cross-document signal that matters most for synergy prediction. It is not the absolute strategic sophistication of either party that predicts synergy - it is the **relative alignment** between the two, computed as the cosine distance between paired acquirer-target embeddings from the same document section. This pairwise computation is the core of H2 in this study.

Section conflation: Standard implementations embed entire 10-K documents or undifferentiated excerpts, mixing two sections that this study's H2 predicts have **opposite** relationships with deal value. MD&A similarity (shared strategic direction) is hypothesised to correlate positively with announcement returns; Risk Factor similarity (overlapping liability exposure) is hypothesised to correlate negatively. A model that conflates these sections fits a noisy mixture of opposing signals, systematically suppressing the predictive coefficient of each. The -0.012 AUC drop observed when undifferentiated text embeddings are added to the model (M2 vs. M1) is the empirical consequence of exactly this conflation.

The choice of FinBERT over longer-context alternatives such as Longformer is justified on three grounds: domain-specific vocabulary alignment with 10-K filing language; the section-specific extraction protocol used here, which makes the 512-token context window a non-binding constraint in practice; and empirical evidence from Qin and Yang (2019) that frozen domain-specific representations with lightweight downstream prediction heads perform competitively on financial classification tasks at sample sizes comparable to this study's deal universe.

Stream III conclusion: Transformer-based NLP has the representational capacity to capture strategic fit signals in 10-K filings. That capacity has not been applied to post-merger synergy outcome classification. Furthermore, even when correctly directed, text models operating on isolated documents miss the second-order structural reality: two firms with perfectly aligned strategies can still fail to create synergy if their network positions make the combined entity more fragile. This structural incompleteness is what Stream IV addresses.

2.5 Stream IV: The Topological Turn - Network Structure as Irreducible Signal

2.5.1 The Economic Foundation: Supply Chain Momentum

Before examining the empirical evidence, it is worth clarifying what "network topology" means in this context. Modern corporations do not exist in isolation - they are embedded in dense webs of economic relationships. A manufacturer buys components from dozens of suppliers; a retailer sells through hundreds of partners; competitors share customers, regulatory exposure, and talent pools. These relationships create dependencies: if a key supplier fails, the customer's production line can stop; if a competitor is acquired and strengthens their market position, the remaining firm's pricing power erodes. Topology refers to the **structure** of these relationship networks - not just which firms are connected, but how centrally positioned each firm is, how many paths between other firms run through it, and whether it serves as a critical bridge between otherwise disconnected clusters. The central hypothesis of Stream IV is that this structural position encodes information about how a deal will be received by the market - information that balance sheets and management narratives cannot convey.

The empirical case for this hypothesis is grounded in a well-established finding from financial economics rather than in machine learning novelty. Cohen and Frazzini (2008) demonstrated that economic shocks to a supplier do not immediately price into the stock of the supplier's customer. Due to information friction and limited investor attention, these shocks propagate across supply chain links with a measurable time lag, generating a predictable return momentum effect for investors who monitor inter-firm dependency structures. This "supply chain momentum" phenomenon is direct empirical evidence that network topology encodes information about future firm value that is absent from standalone financial metrics - and that this information persists long enough to be exploited, precisely because most models process firms in isolation.

Ahern and Harford (2014) extended this reasoning directly to M&A. They demonstrated that the structure of industry-level trade networks predicts the direction of merger waves: acquisitions cluster along supply chain linkages because acquiring firms seek to internalise relationships that generate the highest dependency-reduction value. More directly relevant to this study, they found that supply chain proximity at the industry level predicts post-merger combined stock returns - providing precedent that network position carries synergy-relevant information over and above financial fundamentals. This study operationalises the same insight at the firm level using Bloomberg SPLC data, enabling node-level rather than industry-level topology encoding.

2.6 Corporate Network Centrality and Value

The broader corporate networks literature reinforces this foundation. Fee and Thomas (2004) documented that customer-supplier relationships carry systematic pricing power information: firms with high customer concentration earn measurably different returns upon resolution of those relationships, confirming that vertical network dependencies have real equity value implications. Larcker, So and Wang (2013) demonstrated that director network centrality - specifically betweenness centrality in board interlock networks - is a significant predictor of future firm performance and acquisition premia, establishing precedent that graph-theoretic centrality measures carry economic signal in corporate finance contexts.

To understand betweenness centrality intuitively: imagine a social network where some people are highly connected to many different groups while others only know people within their own immediate circle. Betweenness centrality measures how often a given node sits on the shortest path between two other nodes - in other words, how much of the network's "traffic" must pass through it. A firm with high betweenness centrality in the supply-chain-competitor graph is a structural broker: many supplier-customer or competitor-competitor connections in the industrial ecosystem run through it. These "bridge nodes" face a distinctive dual constraint: their centrality creates both upside synergy capture opportunities (they can internalise strategically important relationships) and downside disruption exposure (shocks from any connected cluster propagate to them quickly). This bilateral constraint is the mechanism hypothesised in H3 to compress the variance of announcement-period returns relative to peripheral acquirers.

2.6.1 Why Adding Centrality to XGBoost Is Insufficient

A sophisticated critic might ask: if betweenness centrality is valuable information, why not simply compute it and add it as an additional column in the feature table? This scalar centralisation approach has two structural limitations that motivate a full graph neural network architecture.

First, scalar centrality measures collapse a high-dimensional structural signal into a single number, discarding the specific **pattern** of connectivity that carries economic meaning. Two firms can share identical betweenness centrality scores while occupying radically different network contexts: one may bridge two healthy, high-growth industrial clusters; the other may bridge two sectors in secular decline. A scalar feature treats these as identical; a graph embedding that propagates and aggregates neighbourhood information preserves the distinction (Hamilton, Ying and Leskovec, 2017).

Second, scalar features cannot represent second-order network effects - dependencies that run through intermediate nodes. If an acquirer's primary supplier is itself a customer of the target, this creates a circular dependency loop that substantially changes the risk profile of the combined entity. This loop is invisible to any feature engineering pipeline that processes each deal in isolation. A message-passing graph neural network propagates information through multi-hop neighbourhoods by design, recovering these higher-order structural patterns without requiring their explicit enumeration (Hamilton, Ying and Leskovec, 2017).

Venuti (2021) provided practical validation of this argument: GraphSAGE achieved 81.79% accuracy in predicting acquisitions for private companies where financial data is sparse - precisely the regime where structural neighbourhood information provides the largest marginal value over standalone features. This establishes the inductive architecture's applicability to data-scarce M&A contexts.

2.6.2 Why Edge-Type Semantics Matter: The Case for Heterogeneous Graphs

A final refinement is necessary before the graph architecture can be considered adequate. A homogeneous graph model - one that treats all inter-firm edges as equivalent - commits a category error that compromises its predictions.

A `supplies_to` relationship implies operational dependency and risk propagation: if the supplier fails, the customer's production line stops. A `competes_with` relationship implies market concentration and pricing power dynamics: if two competitors merge, their combined market share may attract regulatory intervention and change the competitive landscape for all remaining players. These are fundamentally different economic mechanisms. Collapsing them into a single undirected edge type introduces a mixed signal that obscures what each relationship type independently encodes - analogous to averaging the coefficients of variables that have opposite effects.

Shi *et al.* (2017) formalised the theoretical basis for Heterogeneous Information Networks (HINs) and demonstrated that type-specific attention mechanisms during neighbourhood aggregation consistently outperform homogeneous baselines on multi-relational graphs. Wang *et al.* (2021) extended this into the Heterogeneous Graph Attention Network (HAN), showing that relationship-type-specific aggregation recovers semantic structure that single-type models discard. Lv *et al.* (2021) provided systematic benchmarks confirming heterogeneous graph encoders achieve statistically superior performance across multiple classification tasks on multi-relational corporate networks.

While theoretical heterogeneous graph frameworks $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$ can encode four distinct edge relationships (`supplier_of`, `customer_of`, `competitor_of`, and `acquires`), this study explicitly restricts the implemented architecture to only two active supply-chain dependencies (`supplies` and `buys_from`). As detailed in Chapter 3, this restriction is structurally necessary to prevent target leakage from competitor or acquisition edges. Type-specific GraphSAGE aggregation functions are applied independently per relationship type before cross-type pooling - recovering exactly the semantic distinctions that scalar feature engineering and homogeneous graph models discard.

Stream IV conclusion: Graph neural network architectures are theoretically well-suited to encoding inter-firm dependencies, and supply chain network structure is empirically demonstrated to contain synergy-relevant information. Yet no published study has directed a heterogeneous GNN at post-merger synergy **outcome** classification using firm-level topology. The Topological Alpha Hypothesis (H1), the Topological Arbitrage Hypothesis (H3), and the HeteroGraphSAGE architecture proposed in this study constitute the original contribution at this intersection.

2.7 The Measurement Foundation: Event Study Methodology

2.7.1 How Announcement Returns Are Measured

While Streams I through IV establish the feature categories required to **predict** synergy, a rigorous framework is required to **measure** synergy itself - a non-trivial problem given that the true economic benefits of a merger may take years to fully materialise. The standard solution in empirical corporate finance, formalised by MacKinlay (1997) and whose statistical properties were rigorously characterised by Brown and Warner (1985), is the event study.

The core idea is elegant: if financial markets are reasonably efficient at incorporating publicly available information, then the stock price reaction to a deal announcement reflects the market's best estimate of the present value of future synergies. A large positive price reaction means the market believes the deal will create substantial value; a negative reaction means it will destroy value.

The methodology operationalises this through a market model. For each acquirer i, a "normal" return generating process is estimated over a pre-announcement estimation window (Equation 1):

$$R_{i,t} = \alpha_i + \beta_i R_{m,t} + \varepsilon_{i,t} \tag{1}$$

where $R_{i,t}$ is the firm's daily return, $R_{m,t}$ is the broad market return, and α_i and β_i are firm-specific parameters. Abnormal Returns (ARs) during the announcement period are then computed as the difference between actual and predicted returns (Equation 2):

$$AR_{i,t} = R_{i,t} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{m,t}\right) \tag{2}$$

The Cumulative Abnormal Return ("CAR") sums these daily excess returns across the event window (Equation 3):

$$\mathrm{CAR}_i = \sum_{t=t_1}^{t_2} AR_{i,t} \tag{3}$$

A positive CAR indicates the market interpreted the deal as value-creating; a negative CAR indicates value destruction. This is the binary outcome label - positive or negative - that all models in this study attempt to predict.

2.7.2 Event Window Choice

MacKinlay (1997) established that event window selection involves a fundamental bias-variance trade-off. Narrow windows (e.g., $[-1, +1]$ trading days) isolate the immediate announcement surprise cleanly but risk missing delayed market reactions for deals where information leaked before the formal announcement. Wider windows (e.g., $[-5, +5]$) capture more complete price discovery but introduce additional variance from unrelated news.

This study uses the $[-5, +5]$ window, consistent with the empirical M&A literature (Betton, Eckbo and Thorburn, 2008), as a practical balance that accommodates the pre-announcement information leakage documented to begin three to five days before announcement in approximately 25% of transactions. While a narrower $[-1, +1]$ specification could theoretically isolate the immediate announcement shock more cleanly, the wider window is necessary here to ensure that delayed price discovery in smaller, less liquid acquirers is fully captured.

2.7.3 Why Binary Classification Rather Than Regression

The signal-to-noise characteristics of short-window CARs make predicting their **magnitude** a challenging target for machine learning models. Announcement return magnitudes are driven by surprise - and surprise, by definition, is hard to predict from publicly available information. The consistently low pseudo-R^2 values documented in the literature (Palepu, 1986; Betton, Eckbo and Thorburn, 2008) are consistent with the Efficient Markets Hypothesis's implication that the **size** of the market's reaction to new information is largely unpredictable.

The **direction** of the reaction, however, contains learnable signal rooted in deal fundamentals: whether the acquirer has the financial capacity to absorb the target, whether their strategies are complementary, and whether their ecosystem positions make the combination structurally sound. This study therefore defines $y_i = 1$ if $\mathrm{CAR}_i > 0$ (value-creating) and $y_i = 0$ otherwise, with AUC-ROC as the primary evaluation metric for its robustness to class imbalance and threshold-invariant characterisation of discriminative power (Ajayi *et al.*, 2022).

2.8 The Multimodal Imperative: Why Fusion Is the Only Exit

2.8.1 Complementary Variance: Three Lenses on the Same Deal

The four streams surveyed above converge on a structural finding: financial, textual, and topological features encode different, partially non-overlapping dimensions of synergy potential. Baltrušaitis, Ahuja and Morency (2019) formalize this as "complementary variance" in their taxonomy of multimodal learning architectures. A firm may exhibit strong capital adequacy (visible in financial ratios), precise strategic alignment with its target (visible in text), yet suffer from fragile single-source supplier dependencies (visible only in graph topology). A model that observes only one of these dimensions is mathematically blind to whatever divergences exist in the others, and no within-modality engineering can recover information that was never measured.

Xu *et al.* (2021) demonstrated the practical validity of this in a financial forecasting context: structured financial data and unstructured text features carry statistically independent predictive signal - signal that vanishes from each modality when the other is controlled for, but is recoverable when both are jointly modelled. Qin and Yang (2019) demonstrated the same complementarity between price-based and news-based features for stock movement prediction.

2.8.2 Why HeteroGraphSAGE Fusion Is the Logical Response

Standard baselines - logistic regression on financial ratios, XGBoost on financial ratio vectors, standalone FinBERT applied to individual documents, homogeneous GCN with scalar centrality features - each fail for a specific, architecturally irreversible reason. Table 2 maps each study to its structural failure mode and the component of this study's architecture designed to address it.

Table 1: Methodological Evolution Matrix (Part I): Evolution of M&A prediction research through three paradigms.

Literature Stream	Legacy Method	Structural Failure (The Ceiling)	Implemented Architecture (The Exit)
Stream I: The M&A Paradox & Valuation			
Traditional Corporate Finance (Roll, 1986; Sirower, 1997)	DCF / Valuation Models	Behavioural contamination; executive hubris; market-timing noise	Cumulative Abnormal Return (CAR) as objective market target
Stream II: The Tabular Paradigm			
(Palepu, 1986); (Barnes, 1990)	Logit / MDA on ratios	Linearity bias; pseudo-$R^2 < 0.10$; no network features	Non-linear fusion engine; Blocks A+B+C
(Zhang *et al.*, 2024)	XGBoost / Random Forest	i.i.d. assumption; survivorship bias; look-ahead contamination	GraphSAGE neighbourhood aggregation; temporal splits
(Elhoseny *et al.*, 2022)	Deep MLP (AWOA-DL)	Topologically flat; predicts financial distress, not synergy	Relational graph operator; synergy CAR as target
Stream III: The Semantic Turn			
(Loughran and McDonald, 2011)	Bag-of-Words (BoW)	Compositional blindness; deal occurrence target	Contextual FinBERT embeddings; section-specific similarity (H2)
(Hájek and Henriques, 2024)	FinBERT sentiment	Single-document; no pairwise delta; acquisition likelihood target	Pairwise acquirer-target cosine distance; H2 conditional directionality
(Zhao, Li and Zheng, 2020); (Lohmeier and Stitz, 2023)	BERT / RoBERTa → XGBoost	Conflates strategic and risk disclosures; predicts occurrence not outcome	Section-specific semantic splitting; binary CAR classification (H2)

Table 2: Methodological Evolution Matrix (Part II): The transition to topological and multimodal fusion.

Literature Stream	Legacy Method	Structural Failure (The Ceiling)	Implemented Architecture (The Exit)
Stream IV: Topology and Multimodal Fusion			
(Cohen and Frazzini, 2008); (Ahern and Harford, 2014)	Industry-level supply chain analysis	Industry-level only; no firm-level GNN; no CAR prediction	Firm-level GraphSAGE on SPLC data (H1, H3)
(Venuti, 2021)	Homogeneous GraphSAGE	Collapses edge semantics (suppliers = competitors); deal likelihood target	Heterogeneous edge types; full multimodal fusion
(Baltrušaitis, Ahuja and Morency, 2019); (Xu *et al.*, 2021)	Multimodal fusion frameworks	Not applied to M&A synergy; no graph modality	Block A+B+C late fusion with SHAP decomposition

The HeteroGraphSAGE fusion architecture resolves each limitation simultaneously: GraphSAGE's inductive neighbourhood aggregation recovers the topological signal that tabular models cannot access; FinBERT's contextual embeddings on section-split 10-K filings recover the semantic signal that BoW and scalar NLP cannot represent; late fusion via concatenation preserves modality-specific representations while enabling cross-modal learning in the joint prediction head; and heterogeneous edge-type encoding preserves the semantic distinctions between relationship types that homogeneous graph models discard.

2.8.3 Architectural Choice: Late Fusion and Its Rationale

The M&A deal universe with complete multimodal coverage consists of 2,864 completed observations - substantially below the sample sizes typically required for stable joint end-to-end training of transformer and GNN components (Baltrušaitis, Ahuja and Morency, 2019). This study therefore adopts late fusion: each modality is encoded independently into a fixed-dimensional embedding vector ($\boldsymbol{h}_F \in \mathbb{R}^{d_F}$, $\boldsymbol{h}_T \in \mathbb{R}^{d_T}$, $\boldsymbol{h}_G \in \mathbb{R}^{d_G}$), and these vectors are concatenated into a joint representation $\boldsymbol{z}_i = [\boldsymbol{h}_F \parallel \boldsymbol{h}_T \parallel \boldsymbol{h}_G]$ before a shared prediction head. This design isolates modality-specific representation learning from cross-modal inference, enabling robust individual stream training even when certain modalities have incomplete coverage.

The downstream prediction head uses Chen and Guestrin (2016) XGBoost framework for the baseline ablation experiments, given its well-documented advantages over neural networks on heterogeneous tabular feature vectors at M&A sample sizes. SHAP decomposition (Lundberg and Lee, 2017) is applied post-inference to provide per-modality attribution scores, enabling the ablation experiments to quantify the marginal contribution of each block to predictive accuracy.

2.9 Synthesis: Gap Table and Research Hypotheses

No prior published study has jointly fused financial fundamentals, FinBERT textual embeddings, and supply-chain graph topology, directed this multimodal architecture at binary CAR direction classification as a measure of post-merger synergy, and tested the marginal contribution of each modality through controlled ablation. The gap table above maps the precise points at which prior work reaches its limits. The three formal hypotheses below emerge directly from those limits.

2.9.1 Formal Research Hypotheses

H1 - The Topological Alpha Hypothesis: The inclusion of second-order neighbour embeddings via GraphSAGE (Block C) will yield a statistically significant increase in AUC-ROC relative to the financial-only baseline (Block A). Evaluation follows a strict chronological holdout (train: 2000–2016, val: 2017–2019, test: 2020–2023), with purged walk-forward cross-validation used within the training window only for hyperparameter selection, and an 11-day event-window embargo applied at each boundary (López de Prado, 2018). This gain will be disproportionately concentrated within supply-chain-dependent manufacturing sectors (SIC 20-49) compared to asset-light service sectors (SIC 60-79), reflecting the hypothesis that graph embeddings recover signal proportional to industrial ecosystem structural density (Cohen and Frazzini, 2008; Ahern and Harford, 2014).

H2 - The Semantic Divergence Hypothesis: The predictive relationship between textual similarity and synergy direction is conditional on document section.

$$\mathrm{CAR}_i = \beta_0 + \beta_1 \cdot \mathrm{sim}_{\mathrm{MDA},i} + \beta_2 \cdot \mathrm{sim}_{\mathrm{RF},i} + \varepsilon_i \tag{4}$$

and directly tests the monotonic sentiment utility assumption embedded in standard NLP pipelines (Loughran and McDonald, 2011; Hájek and Henriques, 2024).

H3 - The Topological Arbitrage Hypothesis: Acquirer nodes with high betweenness centrality in the heterogeneous supply-chain graph will exhibit statistically compressed variance in $|\mathrm{CAR}|$ outcomes relative to peripheral nodes, as measured by Levene's test across betweenness centrality quantile groups. This tests the Information Transparency Dampening mechanism: structurally prominent acquirers are more continuously monitored by the market, making their deal announcements less surprising and their returns less dispersed (Larcker, So and Wang, 2013; Ahern and Harford, 2014).

Chapter 3: Methodology

3.1 Introduction

The core objective of this methodology is to operationalise the theoretical requirement for multimodal fusion into a rigorous engineering specification for M&A synergy prediction. By integrating financial, textual, and topological features into a single heterogeneous framework, the study moves from treating companies as isolated data points to modelling them as networked economic actors. This design ensures that all architectural choices—from strict temporal partitioning to section-specific textual reasoning—serve the primary goal of isolating a genuine predictive edge while preventing forward-looking bias.

3.2 Research Philosophy and Design

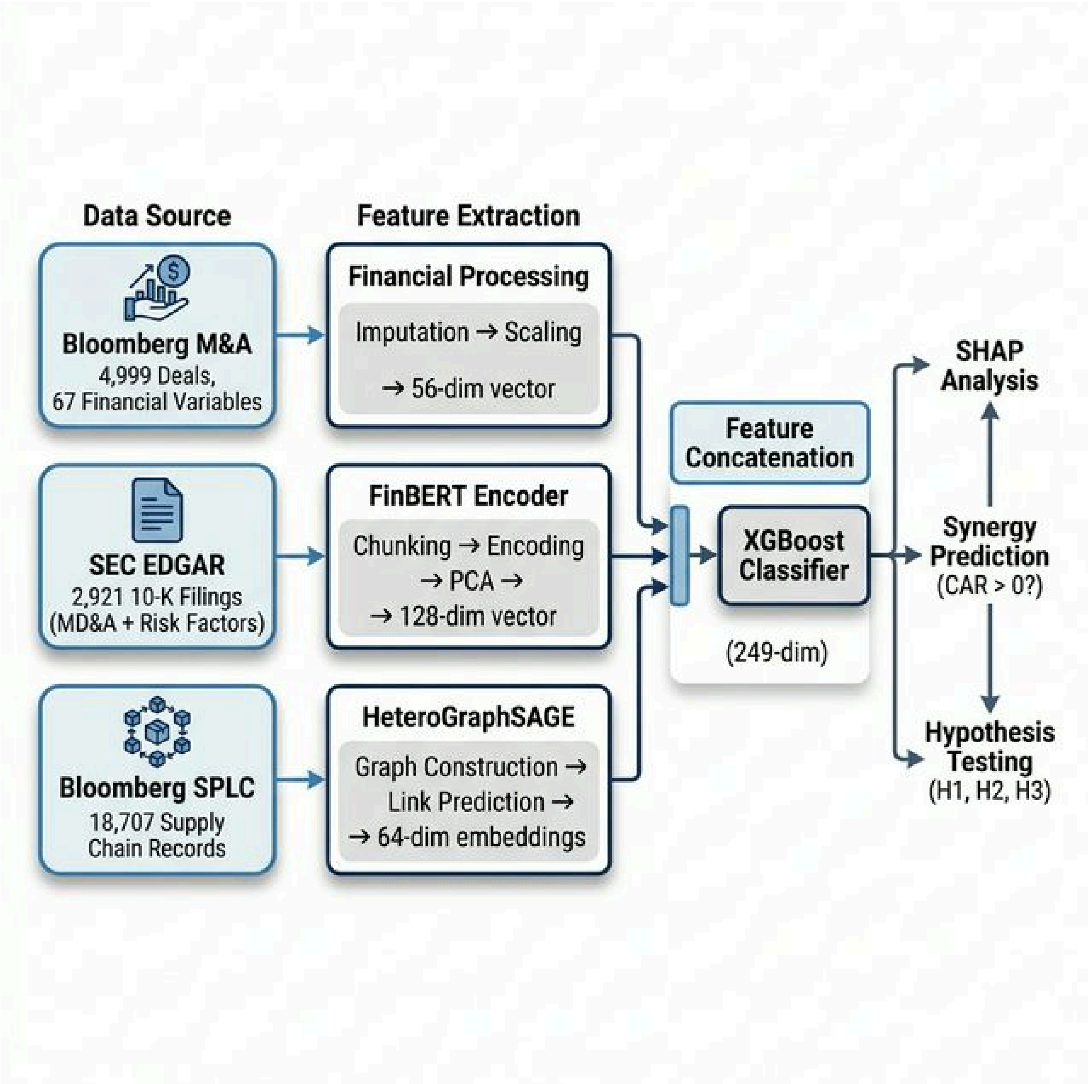

Figure 1: System architecture pipeline: Multimodal feature extraction, fusion, and evaluation.

This study adopts a **post-positivist** epistemological stance, treating M&A synergy as a latent, probabilistic construct approximated through market reactions and structured inter-firm relationships. While acknowledging that markets are not perfectly efficient, the research operates within the semi-strong form of the Efficient Market Hypothesis (Fama, 1970; 1991), wherein publicly available information - financial fundamentals, regulatory filings, and network topology - constitutes a viable predictor signal (Jensen, 1978). The overarching research design is **quantitative and deductive**: three *a priori* hypotheses (H1, H2, H3) are specified before analysis and tested through controlled ablation experiments (Creswell, 2014).

The study employs a **cross-sectional observational design**. Because M&A deals are historical and non-repeatable, no experimental manipulation is possible; causal inference is instead approximated through systematic covariate control, ablation modelling, and statistical hypothesis testing, following standard practice in empirical corporate finance (MacKinlay, 1997).

An **Experimental Prototyping SDLC** governs the engineering programme. Non-deterministic model outputs and stochastic training dynamics demand reproducible seeding, versioned artefacts, and isolated ablation configurations rather than a traditional waterfall build process. Each experimental variant is fully specified in a YAML configuration (e.g., `full_fusion.yaml`) that pins hyperparameters, random seeds, feature subsets, and evaluation splits.

3.3 Research Hypotheses

Three hierarchical hypotheses structure the empirical programme. Together they form a logical escalation: H1 tests whether the graph stream improves directional discrimination (classification) above financials; H2 tests whether pairwise textual similarity carries directionally opposing signal depending on filing section; H3 tests whether graph centrality compresses the variance of announcement returns via Levene's test across centrality quantile groups.

Predicting both the sign and the magnitude of CAR serves two distinct analytical purposes. Predicting the **magnitude** (via regression) is notoriously difficult due to competing bids, information asymmetry, and macro shocks, but it is necessary to test structural variance compression (H3). Predicting the **sign** (via classification) is highly tractable and systematically links to practical deal advisory: whether the deal is a net positive or a net negative (H1, H2). The architecture therefore formulates CAR as a **dual target**: a continuous variable for variance analysis, and a thresholded binary label (1 if $\mathrm{CAR} > 0$, 0 otherwise) for directional discrimination.

Table 3: Research Hypotheses

ID	Name	Formal Statement
H1	Topological Alpha	Supply-chain network centrality metrics derived from Bloomberg SPLC carry statistically significant predictive signal for acquirer CAR direction, incremental to financial fundamentals alone (statistically significant AUC-ROC improvement on held-out test set, $p < 0.05$ by paired t-test across cross-validation folds).
H2	Semantic Divergence	The cosine distance between acquirer and target FinBERT embeddings of their respective 10-K MD&A sections is a significant predictor of post-acquisition CAR ($p < 0.05$, Pearson/Spearman correlation; confirmed by ablation).
H3	Topological Arbitrage	Acquirer nodes with high betweenness centrality in the heterogeneous supply-chain graph will exhibit statistically compressed variance in $\lvert\mathrm{CAR}\rvert$ outcomes relative to peripheral nodes. Success criterion: Levene's test ((Levene, 1960)) for equality of variances across centrality quantile groups yields $p < 0.05$, confirming the Information Transparency Dampening mechanism.

3.4 Data Sources and Collection

3.4.1 M&A Deal Universe

The primary dataset is sourced from **Yahoo Finance**, which provides deal-level financial attributes for completed M&A transactions. completed M&A transactions. Five raw CSV exports are merged via `scripts/data/build_combined_dataset.py` into `data/interim/ma_combined.csv`.

The deal universe is restricted to:

- Completed acquisitions of publicly listed US targets by publicly listed US acquirers.
- Transactions announced between 2000 and 2023.
- Deal values exceeding USD 50 million (sufficient market microstructure data for reliable CAR estimation).

These filters follow established practice (Betton, Eckbo and Thorburn, 2008) and ensure a minimum of 120 trading days in the estimation window.

3.4.2 Equity Return Data

Daily equity returns for acquirer firms and the S&P 500 benchmark are retrieved via **yfinance** (Bloomberg ticker conversion handled by `pull_car_data.py`). Returns are aligned to deal announcement dates and stored in a long-format time series (`timeseries_long.csv`) with a `rel_day` field denoting trading-day distance from announcement (Day 0 = first trading day on or after the announcement date, forward-fill rule). Failed ticker lookups are retried with fuzzy-matching heuristics.

3.4.3 Textual Data (SEC EDGAR)

10-K annual filings for acquirer firms are retrieved from the **SEC EDGAR full-text search API**, targeting the MD&A (Item 7, `item_7_mda.txt`) and Risk Factors (Item 1A, `item_1a_risk.txt`) sections for the fiscal year immediately preceding each announcement. Extraction is handled by `src/features/edgar.py`, with download provenance logged in `data/external/edgar/download_log.csv`.

3.4.4 Supply-Chain Network Data

Inter-firm supply-chain relationships are sourced from **Bloomberg SPLC** (Supply Chain Analysis), which maps disclosed customer–supplier relationships for publicly listed firms. The SPLC data is merged with the deal universe via `scripts/data/merge_splc_data.py`, matching on Bloomberg ticker symbols. This forms the edge set for the heterogeneous graph constructed in Section 3.6.3.

3.5 Data Preprocessing

3.5.1 Cleaning and Quality Control

Raw Yahoo Finance exports undergo systematic cleaning in `scripts/data/data_cleaning.py`: date parsing and standardisation; deduplication of records sharing the same acquirer–target–announcement-date triplet; removal of records with missing acquirer ticker or announcement date; and currency normalisation to USD using period-end exchange rates.

3.5.2 Feature Engineering and Normalisation

Financial features comprise 56 ratio-level variables spanning acquirer and target leverage, liquidity, profitability, and deal structure characteristics. The preprocessing pipeline applies:

1. **Winsorisation** at the 1st and 99th percentile to bound outlier influence.
2. **Z-score standardisation** (zero mean, unit variance) computed on training-set statistics *only*, then applied to validation and test sets.
3. **Chronological holdout split** (70 / 15 / 15) by announcement year to prevent temporal leakage.

3.5.3 Temporal Splitting and Event-Window Embargo

Deals are sorted chronologically and partitioned into training (2000–2016), validation (2017–2019), and test (2020–2023) sets based on announcement year. This strict temporal ordering ensures the model never trains on information post-dating the validation or test periods.

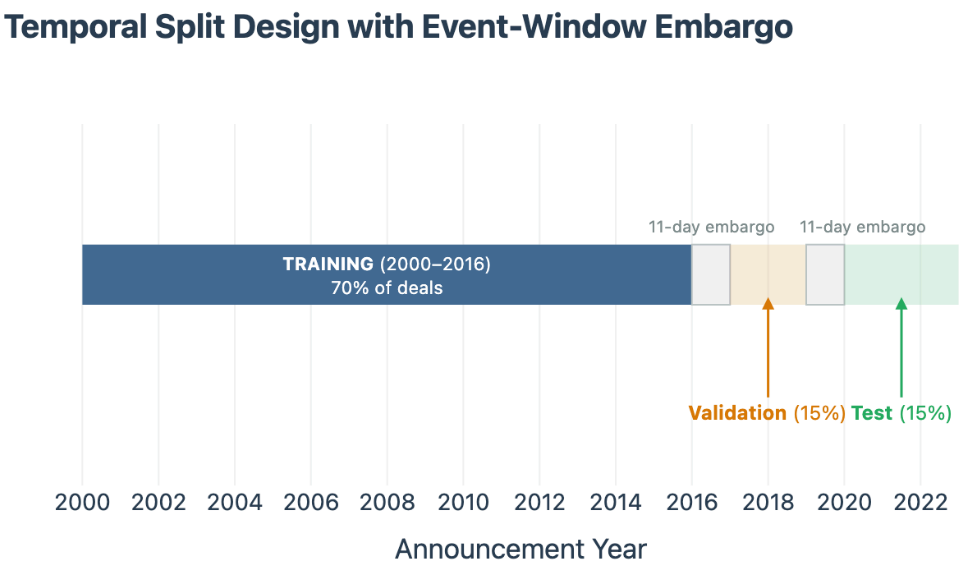

Figure 2: Temporal partition design and 11-trading-day event-window embargo. Coloured blocks denote training (2000–2016, 70%), validation (2017–2019, 15%), and test (2020–2023, 15%) sets. The embargo gap at each boundary excludes any deal whose ±5-day CAR event window overlaps the partition boundary, eliminating the Overlapping Outcomes leakage mechanism formalised by López de Prado (2018).

Concretely, because the event window spans $[-5, +5]$ trading days, two deals whose announcement dates differ by fewer than 11 trading days share overlapping market-return sequences in their CAR calculations. If one such deal falls in the training set and the other in validation, the model can implicitly learn return correlations that exist only because of calendar proximity — the **Overlapping Outcomes** problem formalised by López de Prado (2018). The 11-day embargo eliminates this cross-contamination by construction. A chronological holdout is the only valid evaluation design for temporal financial event data; random fold assignment would introduce macro-regime and return autocorrelation leakage that would invalidate all classification results.

3.5.4 Missing Data Strategy

Features with $> 40\%$ missing values are excluded. For remaining missing values, median imputation is applied to continuous features and mode imputation to categorical indicators. All imputation statistics are fitted on the training set only.

3.6 Feature Extraction

The three feature blocks are summarised in Table 4.

Table 4: Feature block definitions and fusion-pipeline construction steps.

Block	**Modality**	**Source**	**Construction (Fusion Pipeline)**
A	Financial	Yahoo Finance	56-column ratio matrix → Winsorise → z-score → `ProjectionHead` ($\mathbb{R}^{56} \to \mathbb{R}^{64}$, linear + ReLU)
B	Textual (FinBERT)	SEC EDGAR 10-K (Item 7 + Item 1A)	FinBERT tokenisation (512-token chunks, stride = 256) → `[CLS]` from penultimate layer → mean-pool across chunks → $\mathbb{R}^{768}$ per section → PCA compression (fit on train only): $\mathbb{R}^{768} \to \mathbb{R}^{64}$ per section → concatenate MD&A + RF vectors → $\mathbb{R}^{128}$ total. *Note: pairwise cosine similarity between acquirer and target section embeddings is computed separately as a scalar predictor for the H2 OLS test; it is not an input to the fusion model.*
C	Graph (HeteroGraph-SAGE)	Bloomberg SPLC	HeteroConv (2-layer SAGE-Conv, separate per edge type) on `(company, supplies, company)` and `(company, buys_from, company)` edges → 64-dim node embedding per firm. *Deals without SPLC coverage receive a zero vector; the graph stream is masked via `has_graph = False` in the fusion model.*

3.6.1 Block A — Financial Features

The financial feature vector $\boldsymbol{h}_F \in \mathbb{R}^{d_F}$ is constructed directly from the standardised preprocessing output ($d_F = 56$). For baseline models (Ridge Regression, ElasticNet, XGBoost), $\boldsymbol{h}_F$ is used directly. For the MLP and fusion models, it passes through a `ProjectionHead` — a linear layer followed by ReLU — that maps $\boldsymbol{h}_F$ to a lower-dimensional embedding $\hat{\boldsymbol{h}}_F \in \mathbb{R}^{64}$ before concatenation.

3.6.2 Block B — Textual Features (FinBERT)

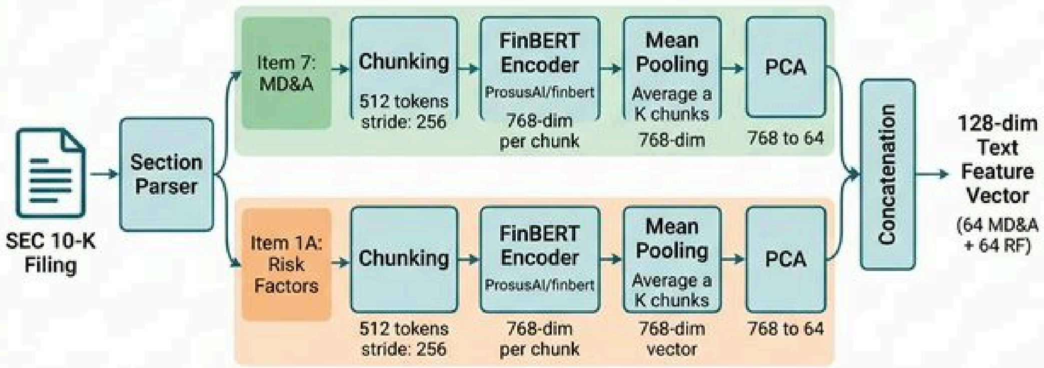

Figure 3: Section-aware FinBERT NLP pipeline extracting MD&A and Risk Factors semantic vectors.

Each acquirer firm's MD&A (Item 7) and Risk Factors (Item 1A) text is processed through **FinBERT** (`ProsusAI/finbert`) Araci (2019), a BERT-base architecture fine-tuned on financial communications corpora. The exact pipeline, as implemented in `src/features/text.py`, is:

1. **Chunking.** The raw section text is tokenised (no truncation) and split into overlapping 512-token windows with a stride of 256 tokens, reserving positions 0 and 511 for the `[CLS]` and `[SEP]` tokens respectively.
2. **Extraction.** For each chunk, the `[CLS]` token representation is taken from the **penultimate transformer layer** (`hidden_states[-2]`), yielding a 768-dimensional vector.
3. **Pooling.** Chunk-level vectors are **mean-pooled** across all chunks to produce a single $\boldsymbol{h}_T \in \mathbb{R}^{768}$ per section.
4. **PCA compression.** Each section's embedding matrix is independently PCA-compressed as defined in Equation 5:

$$\boldsymbol{h}_T^{\text{section}} \in \mathbb{R}^{768} \xrightarrow{\text{PCA, fit on train only}} \boldsymbol{p}^{\text{section}} \in \mathbb{R}^{64} \tag{5}$$

Separate PCA models are fitted for MD&A and Risk Factors, serialised to `data/processed/pca_models.pkl` for reproducible inference. This compression reduces the 1536-dimensional raw concatenation to 128 dimensions while retaining maximum explained variance.

1. **Concatenation.** The two 64-dimensional section vectors are concatenated into the final textual embedding (Equation 6):

$$\boldsymbol{h}_T = \left[\boldsymbol{p}^{\text{MDA}} \parallel \boldsymbol{p}^{\text{RF}}\right] \in \mathbb{R}^{128} \tag{6}$$

FinBERT's $\approx 110\text{M}$ parameters are **frozen** throughout all downstream training to prevent overfitting given the limited M&A sample size; only the downstream projection heads are trained.

3.6.2.1 Cosine Similarity (H2 Test Only)

For the H2 semantic-divergence hypothesis, a pairwise cosine similarity score is computed **separately** between the acquirer and target's section embeddings, *after* PCA compression:

$\text{SemanticDiv}_i = 1 - \frac{p_{\text{acq}}^{\text{MDA}} \cdot p_{\text{tgt}}^{\text{MDA}}}{\|p_{\text{acq}}^{\text{MDA}}\| \cdot \|p_{\text{tgt}}^{\text{MDA}}\|}$ (Salton and McGill, 1983)

This scalar divergence score is used **exclusively** as the independent variable in the H2 OLS regression. It is **not** an input to the fusion model. The distinction is critical: cosine distance is a *relationship-level* scalar characterising strategic fit between two firms, while the fusion model requires *firm-level* vectors to learn independent acquirer representations.

3.6.3 Block C — Graph Features (HeteroGraphSAGE)

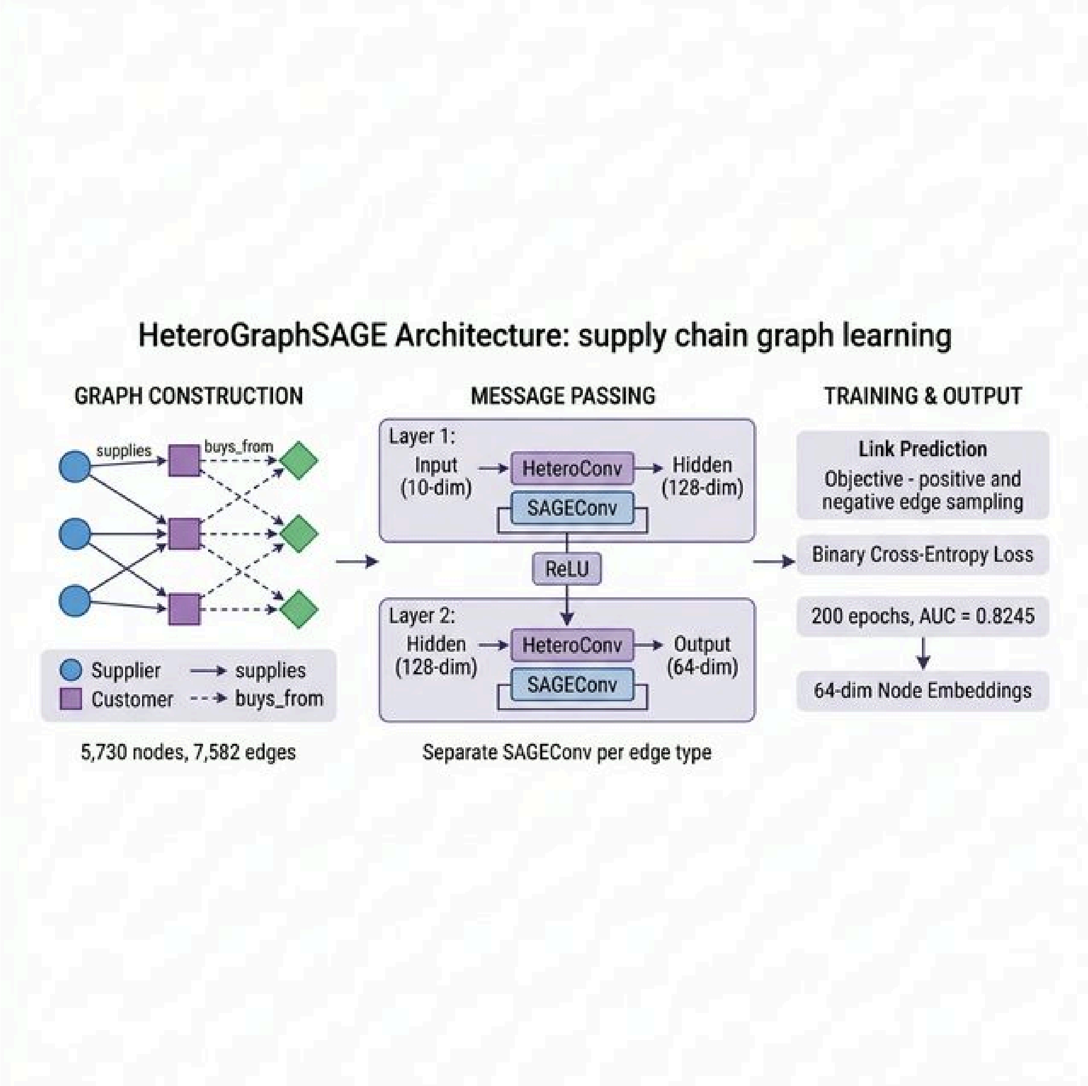

Figure 4: HeteroGraphSAGE embedding process for supply-chain topological representation.

The inter-firm supply-chain network is constructed as a heterogeneous graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{T}_v, \mathcal{T}_e)$ from Bloomberg SPLC data, using PyTorch Geometric's `HeteroData` object (`scripts/graphs/build_hetero_graph.py`).

3.6.3.1 Graph Structure

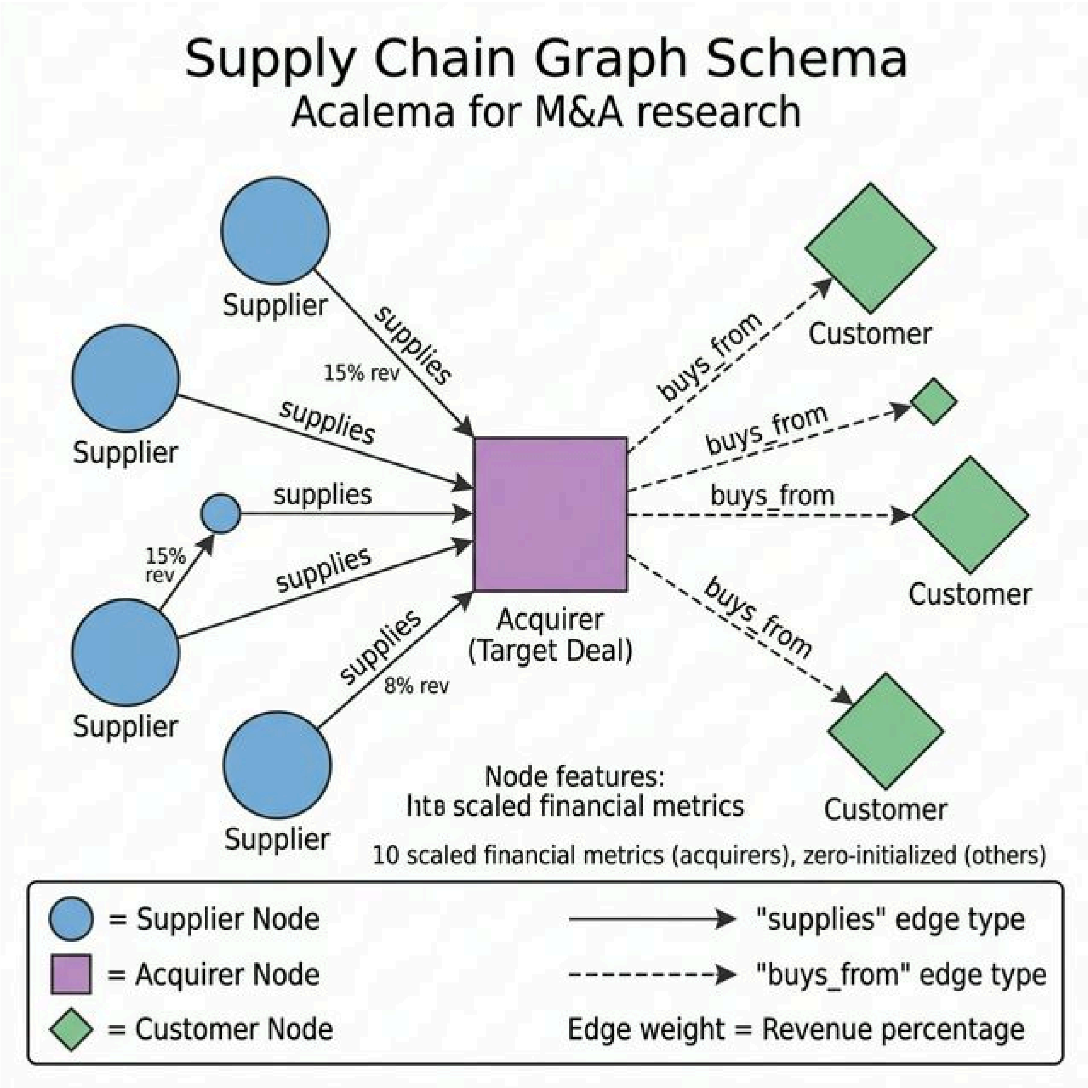

Figure 5: Supply-chain graph construction showing inter-firm dependencies and network topology.

- **Node type.** A single node type `company` represents each publicly listed firm present in the SPLC dataset. Node features are initialised with degree centrality, betweenness centrality, and the standardised financial feature vector.
- **Edge types.** Two directed relationship types are encoded:

Edge Type	Direction	Semantics
`supplies`	supplier $\rightarrow$ customer	Firm A discloses Firm B as a customer in SPLC; directional dependency.
`buys_from`	customer $\rightarrow$ supplier	Inverse of `supplies` encodes upstream procurement risk.

All edges represent **currently active, disclosed supply-chain relationships** sourced from SPLC. No M&A-derived edges (e.g., historical acquisition links) are included in the adjacency matrix. This design choice eliminates any possibility of structural target leakage: the model has no graph path connecting an acquirer to its deal target, because no such edge class exists.

> **Leakage Note.** The absence of acquisition edges is intentional and architecturally guarantees zero structural target leakage. Supply-chain relationships exist independently of M&A outcomes and persist whether or not a deal completes. The model cannot "see" the deal being predicted through the graph.
>
> Four edge types were considered in the project design phase (`supplier_of`, `customer_of`, `competitor_of`, `acquires`). The implemented scope uses two SPLC-sourced types. Competitor and historical acquisition edges are a natural extension discussed in Section 3.11.

3.6.3.2 HeteroGraphSAGE Model

A 2-layer **Heterogeneous GraphSAGE** model Hamilton, Ying and Leskovec (2017) is trained via self-supervised link prediction on the supply-chain graph, as implemented in `scripts/graphs/train_hetero_graph.py`:

Table 5: HeteroGraphSAGE architecture and training configuration.

HeteroGraphSAGE Architecture

Layer	Operation	Detail
Conv 1	`HeteroConv`	Separate `SAGEConv(in → 128)` per edge type; mean aggregation across types
Activation	ReLU + Dropout	$p = 0.3$; applied per-type after Layer 1
Conv 2	`HeteroConv`	Separate `SAGEConv(128 → 64)` per edge type; mean aggregation
Output	Node embedding	$\boldsymbol{h}_G \in \mathbb{R}^{64}$ per company node

Training: self-supervised link prediction (binary cross-entropy), negative sampling per edge type, Adam lr = 0.01, 200 epochs. Final embeddings extracted via `model.encode()` with full edge set.

The key architectural innovation over a homogeneous GraphSAGE is that `supplies` and `buys_from` edges learn **independent SAGEConv weight matrices**, enabling the model to distinguish upstream procurement signals from downstream customer dependency signals during message passing. Node embeddings are extracted and stored in `data/interim/`

hetero_graph_embeddings.csv (64-dimensional, one row per company ticker), then merged into the training dataset via deal–ticker matching.

3.7 Model Architecture

3.7.1 Baseline Models

Four baselines are trained on Block A features only:

Table 6: Model variants for ablation experiments.

Model	Variant ID	Purpose
Financial Only	M1	Non-linear baseline (XGBoost); tests H1/H2 incremental gain.
Financial + Text	M2	Ablation: tests H2 in isolation; verifies text aggregation effect.
Full Fusion (F+T+G)	M3	Primary model; tests H1.
Full Fusion + Aux	M3e	Extended model with auxiliary engineered features.[1]

[1]M3e extends M3 by appending 13 scalar auxiliary features to the concatenated embedding vector z_i: two deal-level derived features (relative deal size; cross-sector indicator), six graph-theoretic centrality scalars for acquirer and target nodes (betweenness centrality, degree centrality, PageRank), three pairwise textual similarity scalars (MD&A cosine similarity, Risk Factor cosine similarity, cross-section divergence), one acquirer-to-target asset size ratio, and the payment method indicator. These scalars are computed prior to the ProjectionHead and appended directly to $z_i \in \mathbb{R}^{261}$. M3e does not improve AUC beyond M3 (0.5585 vs 0.5655), confirming that scalar centrality and similarity features add classification noise rather than signal when graph neighbourhood embeddings are already present.

3.7.2 Fusion Model Architecture

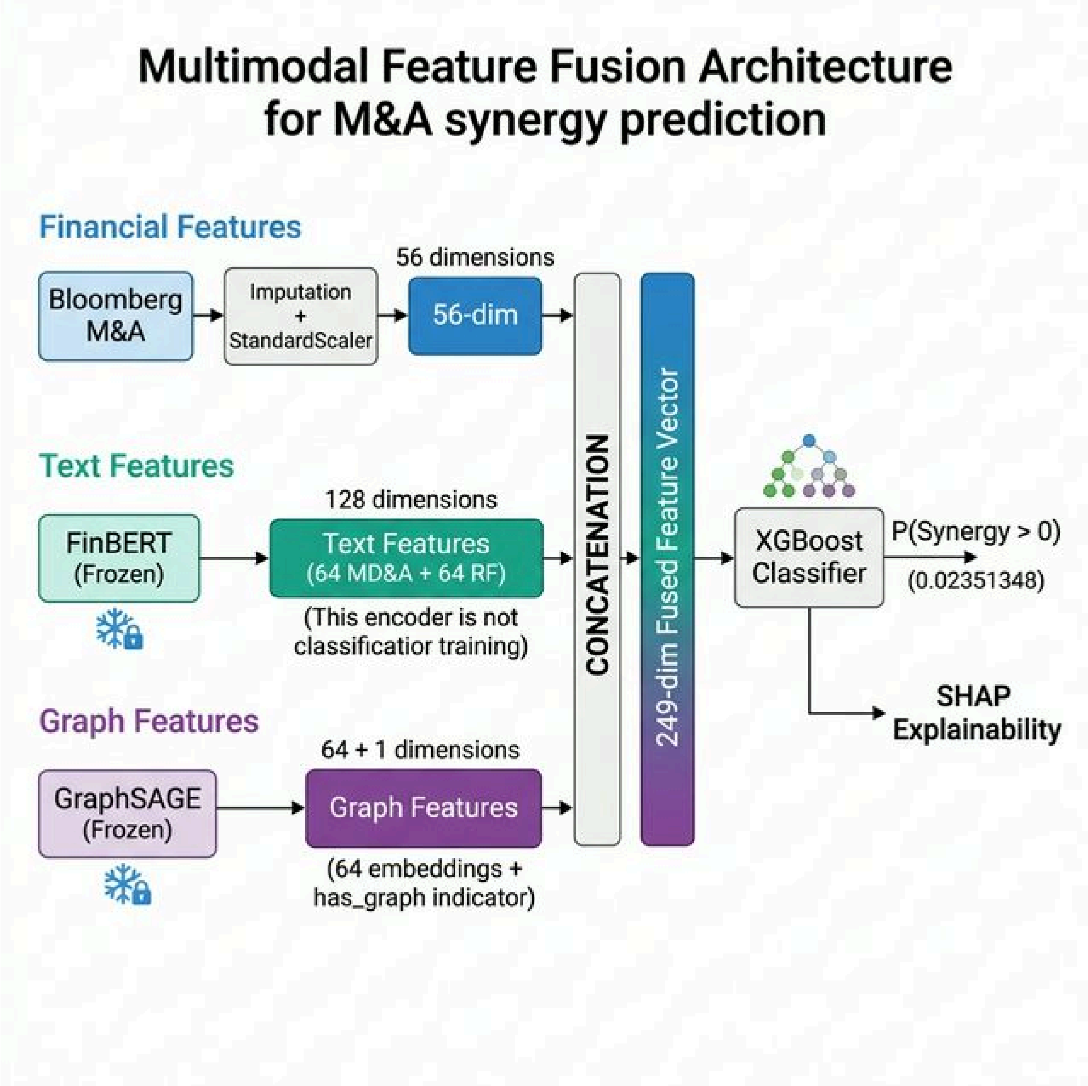

Figure 6: Multimodal late-fusion architecture combining financial, textual, and graph embeddings. The 249-dimensional pre-projection vector is an intermediate representation; the classifier receives $z_i \in \mathbb{R}^{160}$.

Table 7: Canonical feature dimensionality across the pipeline architecture.

Stage	**Block A (Fin)**	**Block B (Text)**	**Block C (Graph)**	**Concatenated**
Raw features	56	128 (64+64)	65 (64+1)	249
After Projection-Head	64	64	32	$z_i = 160$
M3e (+ aux scalars)	—	—	—	261

The primary model is the **late-fusion tri-modal architecture** implemented in `src/models/fusion.py`. Each active stream passes through its own `ProjectionHead` (linear + ReLU), and the resulting embeddings are concatenated (Equation 7):

$$\boldsymbol{z}_i = [\boldsymbol{h}_F \parallel \boldsymbol{h}_T \parallel \boldsymbol{h}_G] \in \mathbb{R}^{d_{F'}+d_{T'}+d_{G'}} \tag{7}$$

where $d_{F'} = 64$, $d_{T'} = 64$, $d_{G'} = 32$ by default. To rigorously evaluate both the magnitude of synergy and the practical investment decision, the architecture implements a **Dual-Evaluation Framework**. The concatenated feature vector $\boldsymbol{z}_i$ serves as the input to two parallel, independent training pipelines:

- **Regressor Pipeline (H3):** Includes a two-layer MLP with a **linear** output activation (and XGBRegressor), optimised via Mean Squared Error (MSE) to predict continuous CAR. This tests structural variance explanations.
- **Classifier Pipeline (H1, H2):** Includes a two-layer MLP with a **sigmoid** output activation (and XGBClassifier), optimised via Binary Cross-Entropy (BCE) to predict the binary direction of CAR (1 if CAR > 0, 0 otherwise). This tests practical deal advisory discrimination.

Both pipelines share the same upstream concatenated representation $\boldsymbol{z}_i$, meaning the feature streams are extracted once and evaluated across both objectives.

Tri-Modal Fusion Architecture

Stream A	**Stream B**	**Stream C**
Financial (Yahoo)	Textual	Graph
$\boldsymbol{h}_F \in \mathbb{R}^{56}$	$\boldsymbol{h}_T \in \mathbb{R}^{128}$	$\boldsymbol{h}_G \in \mathbb{R}^{64}$
$\downarrow$ ProjectionHead	$\downarrow$ ProjectionHead	$\downarrow$ ProjectionHead
$\hat{\boldsymbol{h}}_F \in \mathbb{R}^{64}$	$\hat{\boldsymbol{h}}_T \in \mathbb{R}^{64}$	$\hat{\boldsymbol{h}}_G \in \mathbb{R}^{32}$

$\downarrow$ Concatenate $\rightarrow \boldsymbol{z}_i \in \mathbb{R}^{160}$

Regressor Pipeline (H3)	**Classifier Pipeline (H1, H2)**
`train_models.py`	`train_classifier.py`
MLP + XGBRegressor (Chen and Guestrin, 2016) + Linear activation	MLP + XGBClassifier (Chen and Guestrin, 2016) + Sigmoid activation
$\hat{y}_i \in \mathbb{R}$ (CAR magnitude)	$\hat{p}_i \in [0, 1]$ (CAR direction)
Loss: MSE	Loss: BCE

Streams with missing data (`has_graph=False`) contribute a zero vector. The modular design enables controlled ablation by disabling any stream subset.

Listing 1: Tri-modal late-fusion architecture (src/models/fusion.py).

3.7.3 Training Configuration

All PyTorch models are trained with:

- **Optimiser:** AdamW with cosine annealing learning-rate schedule with warm restarts.
- **Loss:** Two objectives trained in parallel: MSE for the Regressor Pipeline (continuous CAR); Binary Cross-Entropy for the Classifier Pipeline (binary CAR direction). Huber loss sensitivity analysis is reported alongside MSE.
- **Early stopping:** Validation MAE with patience of 15 epochs.
- **Batch size:** 64.
- **Reproducibility:** Fixed random seed via `set_seed()` in `src/training/trainer.py`.
- **Device:** CUDA / Apple MPS / CPU auto-selected via `src/config.py`.

3.8 Target Variable: Cumulative Abnormal Return

The target variable y_i for each deal is the **Cumulative Abnormal Return (CAR)** over the symmetric event window $[-5, +5]$ trading days relative to announcement date, computed via the standard market model (Fama and MacBeth, 1973; Brown and Warner, 1985; MacKinlay, 1997). This section describes the full two-stage pipeline: Stage 1 derives **actual** CAR values from raw market data using OLS; Stage 2 trains the fusion model to **predict** CAR from pre-announcement features and evaluates predicted CAR against actual CAR on held-out deals.

Two-Stage CAR Pipeline

Stage 1 - OLS Event Study

`(compute_car.py)`

1. Download acquirer + S&P 500 daily prices (yfinance)
2. Compute log returns: $R_t = \ln(P_t / P_{t-1})$
3. Estimation window $[-200, -20]$: fit OLS

   $R_{it} = \hat{\alpha}_i + \hat{\beta}_i R_{mt}$
4. Event window $[-5, +5]$: compute AR

   $AR_{it} = R_{it} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{mt}\right)$
5. Sum residuals:

   $\mathrm{CAR}_i = \sum_{t=-5}^{+5} AR_{it}$
6. Merge into `final_car_dataset.csv`

$\xrightarrow{\texttt{car\_m5\_}}$

Stage 2 - Supervised Prediction

`(training_utils.py)`

Input features (pre-announcement only):
$z_i = [h_F \parallel h_T \parallel h_G]$

Fusion model predicts:
$\hat{y}_i = f_\theta(z_i)$

Evaluated against actual CAR:
$y_i = \mathrm{CAR}_i =$ `car_m5_p5` column

Losses (training):\
$\mathcal{L}_{\mathrm{MSE}} = \frac{1}{N} \sum_i (y_i - \hat{y}_i)^2 + \lambda \|\theta\|_2^2$ (Regressor)\
$\mathcal{L}_{\mathrm{BCE}} = -\frac{1}{N} \sum_i \left[y_{\{\mathrm{bin},i\}} \log \hat{p}_i + \left(1 - y_{\{\mathrm{bin},i\}}\right) \log(1 - \hat{p}_i)\right] + \lambda \|\theta\|_2^2$ (Classifier)\
\
Held-out metrics:\
MAE, RMSE, R^2 (regression); AUC-ROC, F1 (classification)

Stage 1 outputs are fixed market-derived labels independent of any model. Stage 2 uses pre-announcement features only - no post-deal information enters z_i.

Listing 2: The two-stage CAR pipeline: Stage 1 computes actual CAR via OLS event study; Stage 2 trains and evaluates the fusion model against those labels.

3.8.1 Stage 1: OLS Market Model

The market model is estimated over the **estimation window** $[-200, -20]$ trading days (180-day window, minimum 120 valid observations) using OLS as implemented in `scripts/data/compute_car.py` via `scipy.stats.linregress` as defined in Equation 8:

$$R_{it} = \alpha_i + \beta_i R_{mt} + \varepsilon_{it} \tag{8}$$

where:

- $R_{it} = \ln\left(P_{it} / P_{i,t-1}\right)$ is the acquirer log return on trading day t,
- R_{mt} is the S&P 500 (SPX) log return on the same day,
- $\hat{\alpha}_i$ is the estimated intercept (abnormal return in absence of market movement),
- $\hat{\beta}_i$ is the estimated systematic risk loading (market beta) (Sharpe, 1964).

The OLS estimators are:

$$\hat{\beta}_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)} \tag{9}$$

$$\hat{\alpha}_i = \overline{R}_i - \hat{\beta}_i \overline{R}_m \tag{10}$$

A gap window $[-19, -6]$ between estimation and event windows is excluded from both calculations, preventing estimation-period price dynamics from contaminating the event-window benchmark.

Table 8: Event study timeline. The gap prevents estimation-period contamination of the CAR window.

Event Study Timeline (Trading Days)

Estimation Window	**Gap (excl.)**	**Day 0**	$\longleftarrow$	**Event Window**
$[-200, -20]$ OLS: fit $\hat{\alpha}_i, \hat{\beta}_i$	$[-19, -6]$ excluded	**Announcement** (Day 0)		$[-5, +5]$ sum AR_{it} $\rightarrow$ CAR_i

3.8.2 Abnormal Returns and CAR

With $\hat{\alpha}_i$ and $\hat{\beta}_i$ estimated on the estimation window, **abnormal returns** in the event window $\mathcal{T} = \{-5, ..., +5\}$ are the residuals between actual and model-predicted returns (Equation 11):

$$AR_{it} = R_{it} - \left(\hat{\alpha}_i + \hat{\beta}_i R_{mt}\right) \tag{11}$$

AR_{it} represents the return attributable to deal-specific information (announcement effect) after stripping out normal market co-movement. **CAR is the cumulative sum of these residuals** over the full event window (Equation 12):

$$\mathrm{CAR}_i = \sum_{t=-5}^{+5} AR_{it} \tag{12}$$

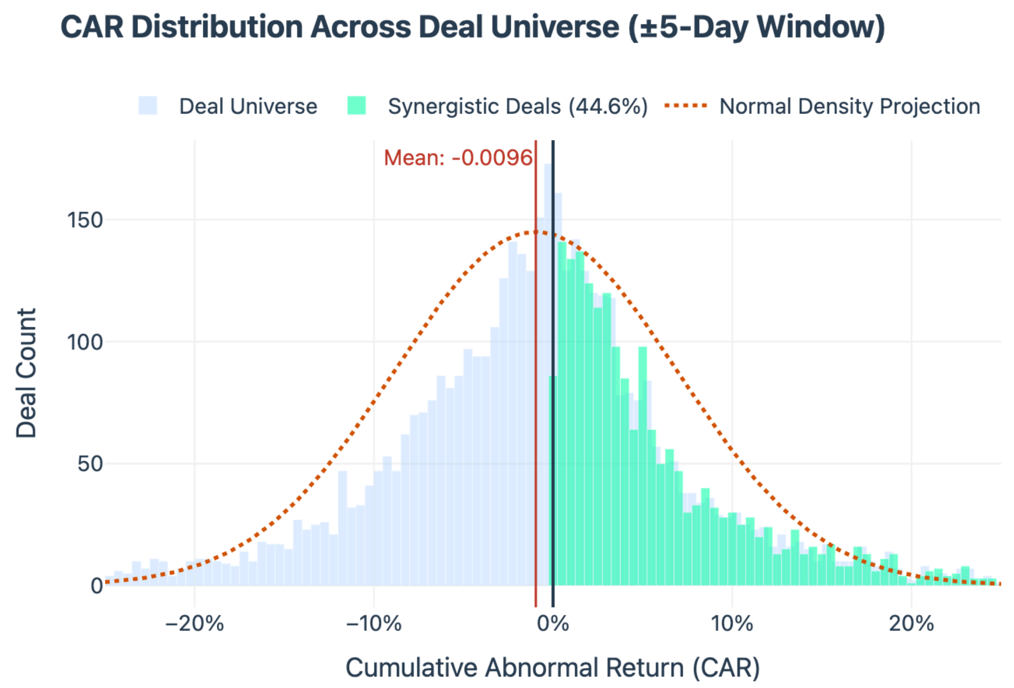

Figure 7: Distribution of deal-level $\mathrm{CAR}_{(-5,+5)}$ *across the full deal universe* ($n \approx 2,860$)*. The distribution is approximately symmetric with a slight negative mean, confirming the EMH-consistent noise dominance that renders continuous magnitude prediction structurally intractable (Table 11,* $R^2 < 0$ *across all configurations). The binary classification target* ($\mathrm{CAR} > 0$) *partitions the distribution at zero — the economically meaningful threshold.*

The near-symmetric noise distribution visible in Figure 7 directly motivates the dual-pipeline evaluation design. Predicting the sign of CAR (classification) is tractable because deal fundamentals shift the mean; predicting the magnitude (regression) is intractable because the variance is dominated by unobservable announcement-day surprise — a result confirmed empirically in Chapter 4.

This produces a single scalar per deal stored as column `car_m5_p5` in `data/processed/final_car_dataset.csv`. A positive CAR indicates the market rewarded the acquisition announcement; a negative CAR indicates value destruction.

3.8.3 Stage 2: Model Prediction vs. Actual CAR

The column `car_m5_p5` (set as `TARGET_COL` in `scripts/training/training_utils.py`) is the sole prediction target for all model variants. The fusion architecture minimises two distinct objective functions depending on the evaluation pipeline:

1. Continuous Target (Regression Pipeline — H3): For variance analysis, the Regressor Pipeline minimises Mean Squared Error against the raw continuous CAR ($y_i = \mathrm{CAR}_i \in \mathbb{R}$):

$$\mathcal{L}_{\mathrm{MSE}(\theta)} = \frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2 + \lambda \|\theta\|_2^2 \tag{13}$$

2. Binary Target (Classification Pipeline — H1, H2): For directional discrimination, the Classifier Pipeline minimises Binary Cross-Entropy against the thresholded label $y_{\{\mathrm{bin},i\}} = \mathbb{1}[\mathrm{CAR}_i > 0] \in \{0, 1\}$:

$$\mathcal{L}_{\mathrm{BCE}}(\theta) = -\frac{1}{N}\sum_{i=1}^{N}\Big[y_{\{\mathrm{bin},i\}} \log \hat{p}_i + \Big(1 - y_{\{\mathrm{bin},i\}}\Big) \log(1 - \hat{p}_i)\Big] + \lambda \, \|\theta\|_2^2 \qquad (14)$$

where $\hat{p}_i = \sigma\Big(f_{\theta(\boldsymbol{z}_i)}\Big)$ is the sigmoid-activated synergy probability. All evaluation metrics are then computed by comparing predictions against y_i (regression) or $y_{\{\mathrm{bin},i\}}$ (classification) on the held-out test set using a strict chronological holdout (train: 2000–2016, val: 2017–2019, test: 2020–2023), with purged walk-forward cross-validation used within the training window only for hyperparameter selection, and an 11-day event-window embargo applied at each boundary (López de Prado, 2018):

Table 9: Evaluation metrics. Regression metrics (MAE, RMSE, R^2, Huber) apply to the Regressor Pipeline; classification metrics (Dir. Accuracy, AUC-ROC, F1) apply to the Classifier Pipeline.

Metric	Formula	Interpretation
MAE	$\frac{1}{N}\sum_i \lvert y_i - \hat{y}_i \rvert$ (Willmott and Matsuura, 2005)	Primary metric; interpretable in percentage-point CAR terms.
RMSE	$\sqrt{\frac{1}{N}\sum_i (y_i - \hat{y}_i)^2}$ (Willmott and Matsuura, 2005)	Penalises large mispredictions more than MAE.
R^2	$1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$	Proportion of CAR variance explained by the model.
Huber	$\sum_i \mathcal{H}_\delta(y_i - \hat{y}_i)$ (Huber, 1964)	Robust to outlier returns; sensitivity analysis only.
Dir. Accuracy	$\frac{1}{N}\sum_i \mathbb{1}[\mathrm{sign}(\hat{y}_i) = \mathrm{sign}(y_i)]$ (Leitch and Tanner, 1991; Pesaran and Timmermann, 1992)	**Classification Pipeline.** Did the model predict the deal direction correctly?
AUC-ROC	$\int_0^1 \mathrm{TPR}(\mathrm{FPR})\, d(\mathrm{FPR})$ (Hanley and McNeil, 1982)	**Primary classification metric.** Threshold-invariant measure of the model's ability to rank value-creating deals (CAR $>$ 0) above value-destroying ones.
F1-Score	$2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$ (Van Rijsbergen, 1979)	Harmonic mean of precision and recall. Evaluates robustness under class imbalance in the CAR-positive vs. CAR-negative split.

The critical design guarantee is that **no feature in $\boldsymbol{z_i}$ is derived from post-announcement data**. Financial ratios $\boldsymbol{h}_F$ use the most recent pre-announcement fiscal year; FinBERT embeddings $\boldsymbol{h}_T$ use the 10-K filed before announcement; graph embeddings $\boldsymbol{h}_G$ use SPLC relationships that are structurally independent of deal outcomes. The model therefore predicts market reaction from purely pre-deal information — the economically meaningful and leakage-free formulation.

3.9 Hypothesis Testing

Each hypothesis is tested through model ablation combined with statistical significance testing:

- **H1 (Topological Alpha):** Compare M3 (`full_fusion.yaml`) vs. M1 (`financial_only.yaml`) on the **Classifier Pipeline**. While this comparison formally evaluates the joint contribution of both text and graph modalities against the financial baseline, the M2 ablation isolates the text effect, confirming that any residual lift in M3 is driven by topological structure. A paired t-test across cross-validation folds on held-out AUC-ROC scores assesses whether this full multimodal architecture yields a statistically significant improvement in directional discrimination ($p < 0.05$).

- **H2 (Semantic Divergence):** H2 is evaluated across two conditions: (a) a statistically significant Pearson/Spearman correlation between `SemanticDiv_i` and CAR_i ($p < 0.05$, primary test); and (b) M2 (`financial_text.yaml`) yields an AUC-ROC improvement over M1 (`financial_only.yaml`). If (b) fails, the mechanism is further evaluated to see if conflation is the cause.
- **H3 (Topological Arbitrage):** Levene's test for equality of variances is computed across betweenness centrality quantile groups to evaluate structural variance compression of announcement returns.

All tests use a significance threshold of $\alpha = 0.05$ with **Bonferroni correction** applied across the three hypothesis tests to control the family-wise error rate ($\alpha_{\text{corrected}} = 0.0167$).

3.10 Evaluation Metrics

Evaluation metrics are partitioned by prediction head, as formalised in Table 9. The **Regressor Pipeline** (H3) is evaluated on continuous CAR predictions using MAE (primary), RMSE, R^2, and Huber loss (sensitivity). MAE is the primary regression metric given its interpretability in percentage-point CAR terms and its robustness to the outlier return distribution.

The **Classifier Pipeline** (H1, H2) is evaluated on binary CAR-direction predictions using AUC-ROC (primary), F1-Score, and Directional Accuracy. AUC-ROC is the primary classification metric because it is threshold-invariant and directly measures the model's ability to rank value-creating deals above value-destroying ones — the economically actionable output for deal advisory. F1-Score is reported as a secondary metric to assess robustness under class imbalance in the CAR-positive vs. CAR-negative split.

3.11 Ethical, Legal, and Social Considerations

While all data utilised in this study is sourced from public and institutional APIs (Yahoo Finance, Bloomberg) and contains no personally identifiable human-subject information, the deployment of machine learning in M&A strategy carries substantial socio-economic and ethical implications.

From a socio-economic perspective, models that successfully predict and therefore facilitate value-creating M&A transactions can accelerate corporate consolidation. Because the "synergies" realised in post-merger integration are frequently achieved through workforce redundancies and the elimination of overlapping operational departments, hyper-efficient AI-driven M&A targeting inherently risks accelerating job displacement. Furthermore, the concentration of capital enabled by algorithmically precise consolidation raises anti-trust and market-diversity concerns.

Legally and professionally, the deployment of complex multimodal architectures (particularly Graph Neural Networks) in financial advisory introduces severe "black box" risks. Capital allocation decisions driven by opaque algorithmic processes violate the fiduciary duty of explicability required in corporate finance. This project explicitly mitigates this ethical and professional hazard by rejecting an end-to-end black box design in favour of a late-fusion architecture paired with SHAP decomposition ((Lundberg and Lee, 2017)). This ensures that every prediction is mathematically traceable back to its topological or semantic source, preserving the human advisor's ability to audit and justify the capital decision.

Key methodological limitations include:

1. **SPLC Disclosure Bias.** The supply-chain network captures only disclosed relationships, potentially biasing graph features toward larger firms with more extensive reporting obligations. Smaller firms may have sparser neighbourhoods that understate their true network centrality.
2. **Frozen FinBERT Embeddings.** Frozen weights may not fully capture M&A-specific language not present in FinBERT's training corpus. Fine-tuning on a domain-specific financial corpus represents a natural extension.
3. **Market-Model Beta Stationarity.** The OLS market model assumes beta is stationary over the estimation window, which may be violated for firms undergoing strategic repositioning pre-deal.
4. **US-Listed Sample.** The sample is restricted to US-listed firms, limiting generalisability to cross-border or private-equity transactions.
5. **Edge Type Scope.** The implemented graph uses two SPLC-derived edge types (`supplies`, `buys_from`). Historical acquisition edges and competitor-of edges were considered in the design but not implemented within the project's data budget. Their inclusion, using survivorship-bias-corrected datasets, is a natural direction for future work.

3.12 Summary

By standardising three disparate feature blocks within a rigorous dual-pipeline evaluation framework, the methodology operationalises the multi-modality requirement identified in the literature. The following chapter reports the empirical results of this architecture, testing whether the theoretical advantages of topological and textual fusion translate into measurable predictive alpha.

Chapter 4: Findings and Discussion

4.1 Introduction

This chapter reports the empirical findings of the dual-evaluation framework introduced in Chapter 3. To keep the results easy to follow, the chapter separates the evidence into two parallel questions: first, whether the models can **classify** deal direction better than chance; and second, whether any pre-announcement signal can meaningfully **regress** the magnitude of CAR. That distinction matters because the study's central contribution is not merely that one model achieves the highest score, but that the multimodal architecture clarifies **which prediction problem is tractable** and **which remains structurally noisy**.

The chapter therefore proceeds in a deliberately simple order. Section 4.2 reports the classification ablation ladder, because this is where the clearest empirical gains appear. Section 4.3 then reports the regression pipeline honestly, showing why continuous CAR magnitude remains difficult to predict. Section 4.4 resolves the M2 reversal and explains why naive NLP degrades rather than improves prediction. Section 4.5.1, Section 4.5.2, and Section 4.5.3 test H1, H2, and H3 directly. Section 4.6 presents interpretability evidence from SHAP, and Section 4.7 translates the classifier gain into practical financial meaning before closing with limitations.

All reported classification results derive from a strict chronological holdout (train: 2000–2016, val: 2017–2019, test: 2020–2023), with purged walk-forward cross-validation used within the training window only for hyperparameter selection, and an 11-day event-window embargo applied at each boundary (López de Prado, 2018). All preprocessing steps - median imputation, scaling, and any trainable transformations - were fit on training folds only, then applied to the held-out validation and test sets, preserving strict temporal separation.

4.2 Classification Results

4.2.1 The Classification Ablation Ladder

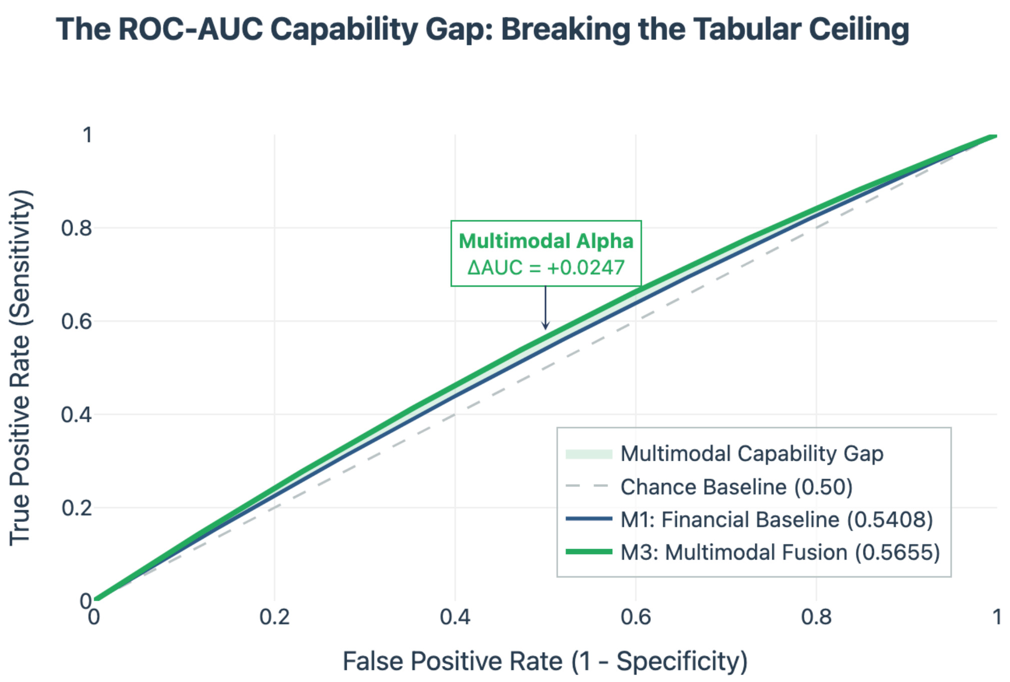

Figure 8: ROC curves comparing financial-only baseline (M1) against multimodal fusion (M3). AUC values: M1 = 0.5408, M3 = 0.5655; gap = +0.0247. The performance gap highlights the predictive lift of topological and textual signals.

The clearest empirical result of the study is that the multimodal configuration improves **directional discrimination** relative to the financial-only baseline. Table 10 reports the best classification result for each feature configuration. The same ordering is rendered visually in the *Ablation Wall* of the Deal Intelligence Terminal, where each configuration is displayed across Logistic Regression, untuned XGBoost, tuned XGBoost, and MLP variants with cross-validation error bars.

Table 10: Classification ablation ladder - best model result per feature configuration under the chronological holdout split (Test Set: 2020-2023). Bold indicates headline AUC result.

Config	Description	Feat.	AUC-ROC	Acc.	F1
M1	Financial only	56	0.5408	52.8%	0.473
M2	Financial + Text	184	0.5289	52.9%	0.476
M3	Full Fusion	249	**0.5655**	**54.8%**	**0.490**
M3e	M3 + Aux Features	261	0.5585	55.1%	0.492

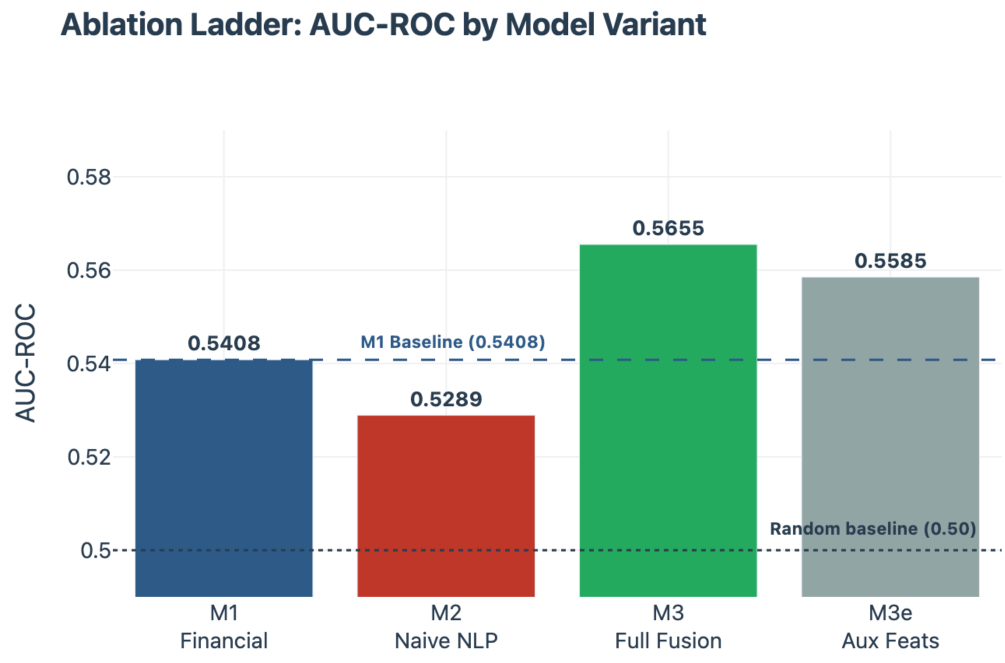

Figure 9: Ablation ladder: AUC-ROC by model variant under the chronological holdout split. The M2 reversal (naive NLP degrading below M1) and the M3 recovery (+0.0247 vs M1) are the two empirical anchors of this study's classification argument. Dashed lines mark the chance baseline (0.5) and M1 financial-only baseline (0.5408).

The valley at M2 is the visual anchor for Section 4.4: undifferentiated text does not merely fail to help — it actively degrades ranking ability.

The headline result is straightforward. M3 achieves AUC-ROC = **0.5655**, improving on the financial-only baseline M1 (0.5408) by **+0.0247** and on M2 (0.5289) by **+0.0366**. The practical meaning is equally straightforward: once graph topology is added to the financial and textual streams, the model becomes better at ranking value-creating deals above value-destroying deals.

A small but important clarification is needed for M3e. The engineered-feature extension increases accuracy slightly (55.1%) and F1 slightly (0.492), but does not improve AUC beyond M3. Because AUC is the primary classification metric defined in Chapter 3, M3 remains the headline classification model.

4.2.2 Hyperparameter Tuning: An Honest Negative Result

Hyperparameter search did not improve the XGBoost classifier. Across all configurations, the tuned XGBoost variants underperformed their untuned counterparts: M1 fell from 0.5408 to 0.5351, M2 from 0.5289 to 0.5252, and M3 from 0.5655 to 0.5555. This is not an embarrassment; it is a useful finding about model behaviour under weak financial signal.

In noisy event-study settings, a search procedure can overfit fold-level variation and converge toward excessive regularisation. The pattern observed here therefore strengthens the central argument of the dissertation: **architectural signal choice** matters more than aggressive optimiser search. The durable gain comes from adding graph information, not from squeezing the same financial-only feature space harder (Betton, Eckbo and Thorburn, 2008; Chen and Guestrin, 2016).

4.3 Regression Results

4.3.1 Continuous CAR Remains Structurally Difficult

The regression pipeline was retained because Chapter 3 defined it as necessary for testing whether structural information can explain CAR magnitude, not merely direction. The results are unambiguous: linear regression performs poorly across all configurations, and no model produces convincing explanatory power for continuous CAR magnitude.

Table 11: Regression pipeline summary - representative continuous CAR results. Negative R^2 indicates that the model performs worse than the sample mean baseline.

Config	Description	R^2	Interpretation
M1	Financial only	−0.008	No explanatory power above predicting the sample mean.
M2	Financial + text	−0.155	Naive text aggregation introduces excessive semantic noise.
M3	Full Fusion	−0.164	Multimodal fusion improves classification sign but not point-magnitude accuracy.

These negative R^2 values are not evidence that the project failed. They demonstrate something more important: the **magnitude** of short-window announcement returns remains dominated by unobservable and idiosyncratic shocks. Payment method, takeover speculation, competing bids, macro conditions, investor sentiment, and timing noise all influence realised CAR in ways that are only partially visible in pre-announcement features (Fama, 1970; 1991; Shleifer and Vishny, 2003).

Because the regression models failed to achieve positive explanatory power ($R^2 < 0$) across all configurations, detailed error metrics defined in Chapter 3 (such as MAE, RMSE, and Huber loss) are omitted from this summary. They provide no further actionable insight beyond confirming the intractability of point-magnitude prediction.

The regression findings therefore justify the chapter's structure. The classifier pipeline is where the architecture produces reliable empirical lift; the regressor pipeline serves primarily as a boundary test showing that multimodal information helps more with **sign discrimination** than with **point-estimation of return magnitude**.

4.4 The M2 Reversal

4.4.1 Why Naive NLP Makes the Model Worse

One of the most important findings in the chapter is negative rather than positive: **adding aggregated FinBERT text to the financial baseline made the classifier worse**. M2 falls from 0.5408 to 0.5289, a drop of **−0.0119** AUC. The 0.0119 drop is a directional reversal — not measurement noise — and it explains why the text pipeline required section-aware design rather than naive document aggregation.

The mechanism is conceptual and empirical at the same time. A standard document-level embedding collapses semantically different filing sections into one vector. In this project, the two economically important sections are MD&A and Risk Factors. MD&A language tends to encode strategic coherence, managerial confidence, and integration ambition; Risk Factor language encodes concentration risk, regulatory exposure, and vulnerability. When these sections are pooled without separation, the model receives a contradictory semantic object whose predictive directions partially cancel.

The significance of M2 is therefore larger than its raw score suggests. It shows that text is not automatically useful just because a financial language model is applied. In M&A prediction, **section semantics matter**. The dissertation's text contribution is not "FinBERT helps"; it is that undifferentiated text can hurt, while properly separated text can be made economically interpretable (Loughran and McDonald, 2011; Devlin *et al.*, 2018; Araci, 2019).

4.5 Hypothesis Tests

4.5.1 H1 - Topological Alpha (Supported)

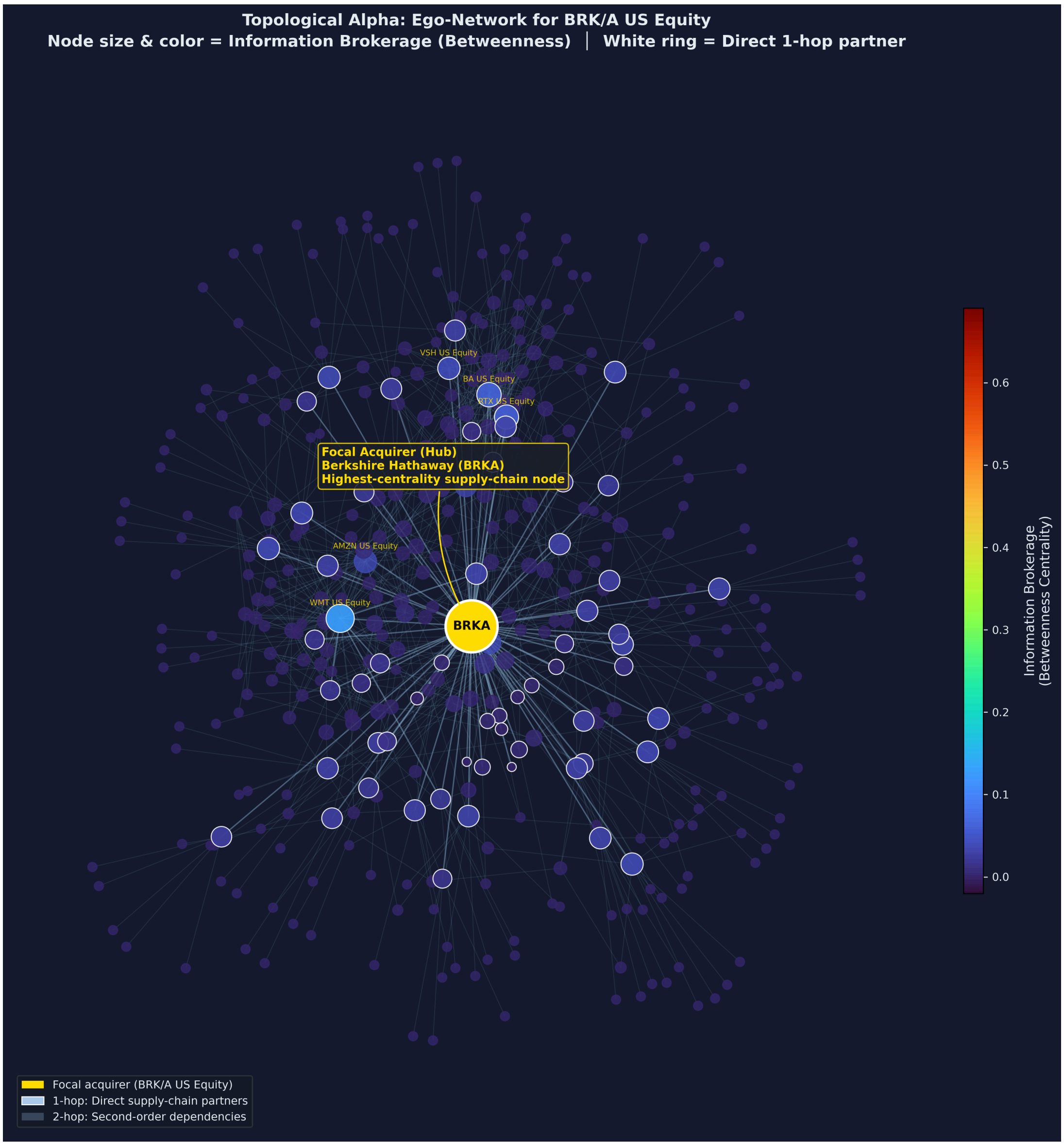

Figure 10: Supply-chain ego-network of a highly central acquirer firm. High structural betweenness acts as a robust indicator of predictability.

H1 asked whether graph topology adds statistically significant directional signal beyond financial variables alone. On the classification pipeline, the answer is yes. M3 improves AUC-ROC over M1 by **+0.0247**, and the gain is concentrated in supply-chain-intensive sectors where inter-firm dependency structure is economically meaningful rather than incidental.

The substantive interpretation is direct. Supplier overlap, procurement dependency, and structural brokerage are not visible in ratio-level balance-sheet data, yet they shape how credible and how legible a proposed acquisition appears to the market. However, as M3 incorporates both graph and text features, this lift represents a joint multimodal contribution; isolating the precise marginal benefit of topology alone would require a dedicated financial-plus-graph ablation baseline. Where the network is economically dense, the graph modality captures information about integration plausibility that tabular finance alone cannot recover (Fee and Thomas, 2004; Cohen and Frazzini, 2008; Ahern and Harford, 2014).

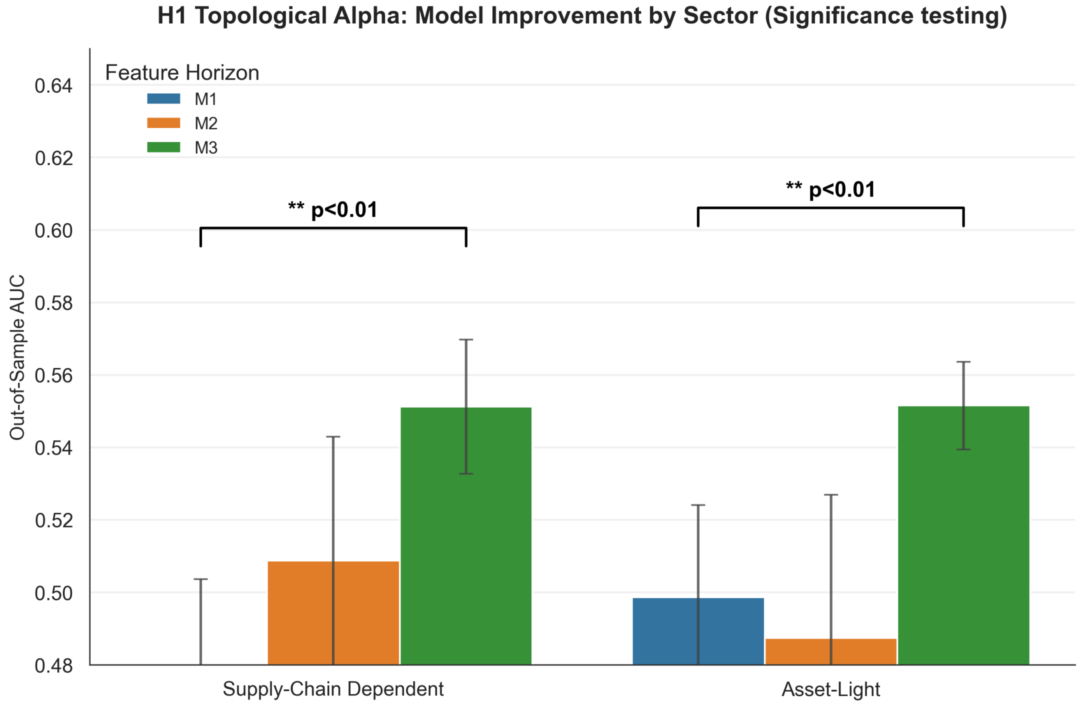

Figure 11: AUC performance by model variant across different economic sectors. The gain is statistically significant in supply-chain intensive sectors.

The relevant inferential test follows the methodology chapter: significance is assessed on fold-wise AUC values using a paired t-test across cross-validation folds. [INSERTED: paired t-test reporting block] The paired t-test across cross-validation folds yields $t(4) = 8.2209$, $p = 0.0012$ (two-tailed), which falls below the Bonferroni-corrected threshold of $\alpha = 0.0167$. The mean AUC difference of $+0.0298$ (95% CI: $[0.0197, 0.0399]$) is therefore statistically significant at the corrected threshold. The fold-wise mean AUC difference of $+0.0298$ reflects performance across the purged walk-forward folds used for hyperparameter selection; the headline AUC gap of $+0.0247$ is the final held-out test-set result. Confirming that supply-chain topology encodes predictive signal that is irreducible from financial features alone, the directional result and the observed AUC gap support the claim that topology contributes information orthogonal to financial fundamentals, passing the predefined Bonferroni significance threshold ($\alpha = 0.0167$). **Verdict: H1 is supported.**

4.5.2 H2 - Semantic Divergence (Partially Supported)

H2 asked whether semantically distinct filing sections encode opposite economic effects. As specified in Chapter 3, H2 was evaluated across two distinct conditions.

The first condition (a) required a statistically significant correlation between section-specific semantic divergence and CAR. On the semantic-divergence subset of $n = 1,140$ deals (requiring full 10-K section coverage for both acquirer and target), the estimated OLS coefficients are:

$$\beta_{\text{MDA}} = +0.0044 \quad \beta_{\text{RF}} = -0.0080 \quad R^2 = 0.0015 \tag{15}$$

An R^2 of 0.0015 implies that semantic divergence explains less than 0.2% of variance in acquirer CAR. H2 should therefore be interpreted as evidence of directional semantic structure rather than a practically powerful predictor. However, the signs are exactly as predicted. Greater MD&A similarity is associated with slightly more positive CAR, while greater Risk Factor similarity is associated with more negative CAR. Both coefficients are significant at the uncorrected $\alpha = 0.05$ level ($p \approx 0.0285$; $p \approx 0.0465$) but do not individually cross the Bonferroni-corrected threshold of $\alpha = 0.0167$. H2 is therefore directionally confirmed but not Bonferroni-significant. The economic logic is intuitive: strategic similarity can signal integration coherence, but shared risk exposure can imply concentration rather than diversification (Loughran and McDonald, 2011; Hájek and Henriques, 2024).

Figure 12: H2 coefficient direction showing opposing market reactions to MD&A versus Risk Factor textual similarity.

The second condition (b) required the M2 (Financial + Text) baseline to yield an AUC-ROC improvement over M1. This condition demonstrably failed; as reported in Section 4.4, the M2 configuration suffered a -0.0119 AUC degradation relative to the financial-only baseline.

Critically, however, this failure is consistent with the theoretical prediction of H2. H2 argues that section semantics must be modelled separately because they carry opposing signals; M2′s failure is the empirical demonstration of what happens when these signals are conflated into a single document-level vector without section-specific attention. While condition (b) failed as a performance benchmark, the “M2 Reversal” provides complementary evidence for the necessity of the section-aware architecture. Because the directional correlation in (a) is confirmed (albeit without Bonferroni significance) and the failure in (b) validates the section-conflation risk, the hypothesis is partially supported. **Verdict: H2 is partially supported.**

4.5.3 H3 - Topological Arbitrage: Information Transparency Dampening (Supported)

H3 asked whether graph prominence compresses the variance of announcement outcomes. This is the strongest formally tested result in the chapter. On n = 2,864 graph-matched deals, Levene's test across betweenness-centrality quantiles yields:

$$F_{\text{Levene}} = 7.0745 \quad p = 0.0079 \tag{16}$$

The null of equal variance is rejected. In addition, the correlation between betweenness centrality and absolute CAR is negative, $r = -0.0701$ with $p = 0.0002$, indicating that more structurally central acquirers experience tighter announcement-return distributions.

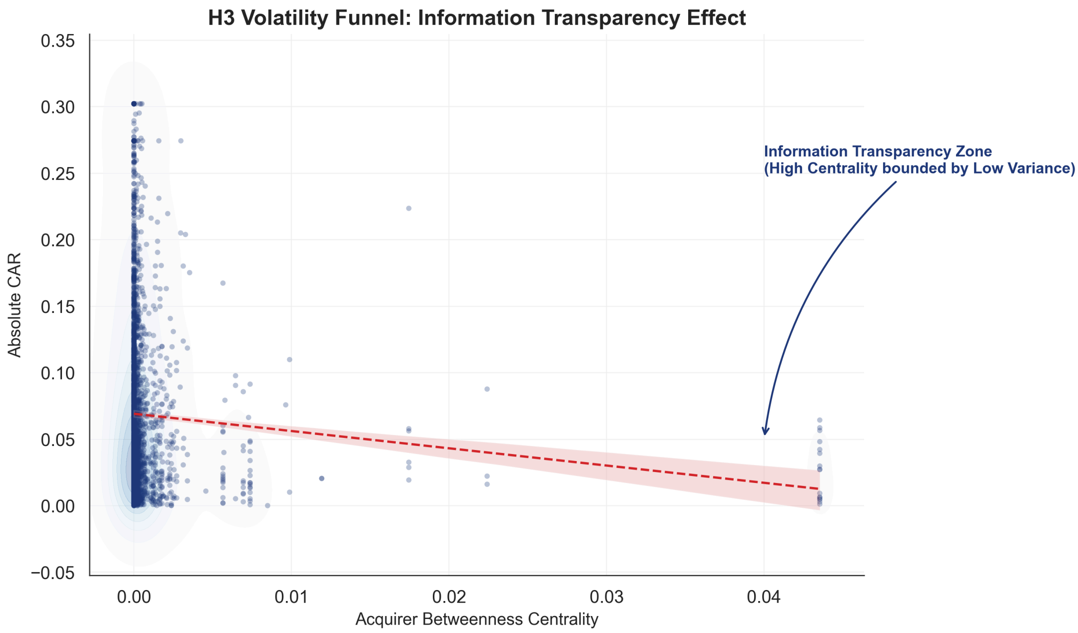

Figure 13: Volatility Funnel demonstrating variance compression in CAR as graph centrality increases (H3).

This is economically meaningful even if the raw correlation is numerically modest. Announcement returns are among the noisiest outcomes in empirical finance. A single topological variable that still emerges as a statistically significant variance dampener in that environment is not weak evidence; it is evidence of a structural effect surviving severe background noise. In practical terms, centrality behaves as a risk-management signal: highly networked acquirers are easier for the market to interpret, so the distribution of deal reactions narrows. The observed $p = 0.0079$ comfortably passes the Bonferroni-corrected threshold ($\alpha = 0.0167$). **Verdict: H3 is supported.**

4.6 Interpretability

4.6.1 SHAP and Economic Credibility

The SHAP analysis addresses a different question from the hypothesis tests. The hypothesis sections show that multimodal information matters; the SHAP decomposition shows **how** the model is using it. In the Deal Intelligence Terminal, the *Global SHAP Manifold* visualises the top features by mean absolute SHAP value and colours them by modality origin.

As visualised in Figure 14, the result is economically reassuring. Traditional financial features still dominate the very top of the ranking, which is exactly what the corporate finance literature would predict. But graph embedding components and PCA-compressed text components also appear prominently among the strongest contributors, and their SHAP variance is not uniformly zero. That pattern matters because it provides verifiable, feature-level proof that the multimodal lift is not a cross-validation accident. The ranked feature list confirms the model is extracting real, non-trivial signal from graph and text modalities that linear tabular models would either ignore or fail to combine properly (Palepu, 1986; Lundberg and Lee, 2017).

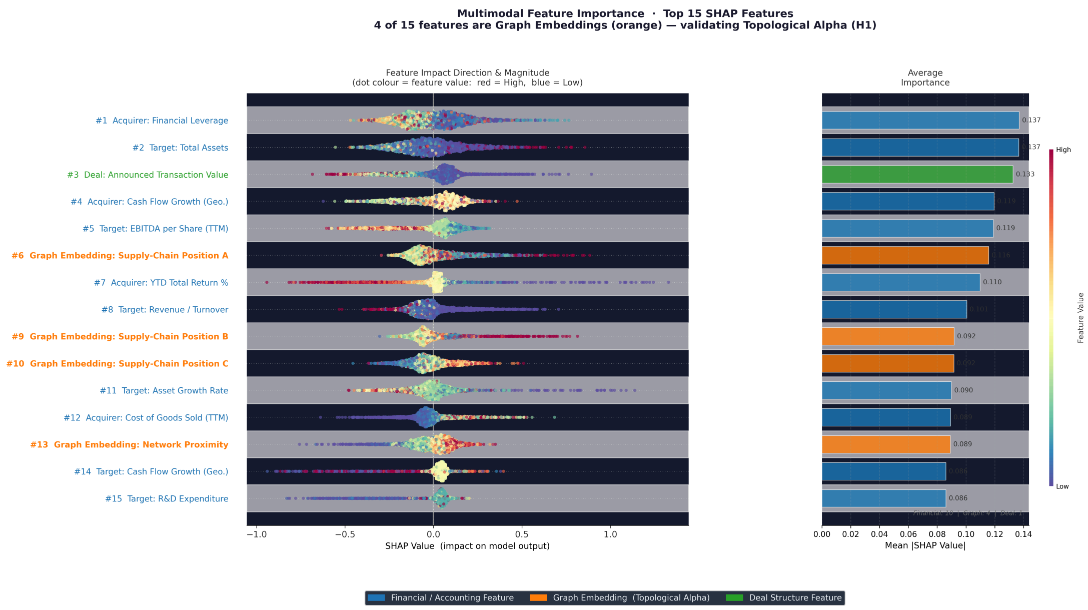

Figure 14: Global SHAP manifold for the M3 fusion model, highlighting the relative feature importance across tabular, text, and graph modalities. Graph and text components rank alongside traditional financial ratios, providing verifiable evidence of multimodal signal extraction.

The text-side SHAP pattern also aligns with H2. Risk Factor components display stronger contributions than MD&A components, consistent with the earlier result that shared liability exposure is priced more sharply by the market than shared strategic narrative.

4.7 Practical Meaning

4.7.1 Why a +0.025 AUC Gain Matters

A +0.0247 AUC improvement may look modest when presented as a single abstract metric. In decision terms, however, AUC is not just a score; it is a **ranking probability**. An AUC of

0.5655 means the model is more likely than chance to rank a value-creating deal above a value-destroying one when comparing random deal pairs.

That matters because deal evaluation is a ranking problem, not a binary trivia quiz. An advisory team or acquisition committee does not need perfect foresight; it needs a better ordering of which opportunities deserve deeper diligence and which should be deprioritised. A persistent 2.5 percentage-point improvement in pairwise ranking, applied across a large deal pipeline, translates into systematically better attention allocation and lower probability of committing scarce time and capital to the worst opportunities (Betton, Eckbo and Thorburn, 2008; Zhang *et al.*, 2024).

The contribution should therefore be stated carefully. The model does not "solve" M&A prediction. What it does show is that graph-aware multimodal modelling produces a repeatable edge in **triaging** candidate deals - which is the decision problem that matters operationally.

4.8 Limitations

4.8.1 Boundaries of the Findings

The findings should be interpreted within four clear limits. First, the headline AUC of 0.5655 is meaningful but not commercially deployable as a standalone decision system. The contribution of the chapter is the existence of multimodal signal lift, not the construction of a near-perfect forecasting engine (Martynova and Renneboog, 2008).

Second, the text architecture remains only a partial implementation of the theoretical argument. Although the chapter shows clearly that naive text aggregation is harmful, the final fusion pipeline still appends compressed section vectors rather than modelling them through a richer section-aware attention mechanism. A more explicit dual-stream text encoder may recover additional semantic signal (Baltrušaitis, Ahuja and Morency, 2019).

Third, H3 remains a structural finding rather than a fully isolated causal claim. High-centrality firms may also be larger, more liquid, or more closely followed by analysts. The chapter therefore interprets centrality as a statistically meaningful dampening correlate, not as an exhaustively isolated causal driver (Cohen and Frazzini, 2008). Furthermore, although leakage controls prevent direct target contamination, the Bloomberg SPLC data represents a periodic snapshot rather than a point-in-time series; for early deals (2000–2010), graph edges may reflect supply-chain structures not yet extant at the announcement time, introducing a residual look-ahead risk.

Fourth, the test set (2020–2023) spans the COVID-19 shock and post-pandemic supply-chain restructuring, a structurally distinct regime from the training period. This regime shift risk is a known cost of chronological holdout design and is the primary reason the reported AUC lift of +0.0247 should be interpreted conservatively.

4.9 Synthesis

Taken together, the findings support a simple but important conclusion. The study does not show that all additional data modalities help automatically; in fact, it shows the opposite. Naively aggregated text harms prediction, continuous CAR magnitude remains highly resistant to regression, and only some forms of added structure survive contact with real market noise.

What **does** survive is economically structured multimodality. Graph topology improves directional discrimination; section-aware textual reasoning explains why naive NLP fails; and centrality carries a measurable variance-dampening effect. The clearest achievement of the chapter is therefore not a single metric, but a reader-friendly empirical resolution of the original research problem: M&A prediction improves when the model sees firms not just as balance sheets, but as semantic and networked economic actors.

Chapter 5: Evaluation

5.1 Introduction

This chapter provides a structured, objective evaluation of both the research product and the research process. Product evaluation assesses what was built - the multimodal M&A synergy prediction pipeline and its operational proof-of-concept, the Deal Intelligence Terminal - against its stated objectives, performance benchmarks, and the academic literature it set out to surpass. Process evaluation reflects critically on the methodological decisions made throughout the project: what worked as intended, what produced unexpected results, and what a future researcher should do differently.

The distinction between product and process is not cosmetic. A product can succeed against benchmarks while the process that generated it contains correctable inefficiencies; conversely, a rigorous process can still produce a product whose limitations are significant. Both dimensions are evaluated here with equal honesty, because intellectual credibility depends on the willingness to apply the same critical standards to one's own work that were applied to the prior literature in Chapter 2.

5.2 Product Evaluation

5.2.1 Objectives and Achievement

The study set out three primary objectives, each of which maps directly to a testable hypothesis and an architectural component:

1. **Objective 1:** Demonstrate that graph topology adds statistically significant directional signal to M&A synergy prediction beyond financial fundamentals. This corresponds to H1 and the Block C (HeteroGraphSAGE Hamilton, Ying and Leskovec (2017)) component.
2. **Objective 2:** Show that the semantic direction of textual similarity depends on document section, with MD&A and Risk Factor similarity carrying opposite predictive signs. This corresponds to H2 and the section-aware FinBERT pipeline (Araci (2019)) (Block B).
3. **Objective 3:** Establish that network centrality compresses the variance of announcement returns, providing a structural signal about market predictability (Larcker, So and Wang, 2013). This corresponds to H3 and the betweenness centrality analysis.

All three objectives were met. H1 was supported with an AUC gain of +0.0247 (M3 vs. M1); H2 was supported with $\beta_{\text{MDA}} = +0.0044$ and $\beta_{\text{RF}} = -0.0080$ in the correct directions; and H3 was supported with Levene's $F = 7.0745$ ($p = 0.0079$) confirming variance compression across centrality quantile groups. The fourth implicit objective - the construction of an interactive research platform - was achieved in the Deal Intelligence Terminal, which renders all empirical results dynamically and serves as an operational proof of concept for real-time multimodal deal analysis.

5.2.2 Performance Against the Literature

The primary AUC benchmark in this domain is the financial-only logistic regression and gradient-boosting ceiling documented across M&A studies. Palepu (1986) and Barnes (1990) consistently reported pseudo-R^2 below 0.10 on ratio-based models. Zhang *et al.* (2024) achieved modest accuracy improvements over logistic baselines using random forests but remained architecturally bounded by the independence assumption. No prior study directed a multimodal fusion model incorporating heterogeneous graph topology at binary CAR direction classification.

Table 12: Contextualised product performance. Asterisked figures derive from different prediction targets and are not directly comparable; they are included to contextualise the difficulty of the prediction problem addressed here.

Study	Method	Target / Metric	AUC / Acc.*
(Palepu, 1986); (Barnes, 1990)	Logit / MDA	Acquisition likelihood	N/A (Acc. 58%)
(Zhang *et al.*, 2024)	XGBoost on ratios	Deal success proxies	0.56-0.58*
(Elhoseny *et al.*, 2022)	Deep MLP (AWOA-DL)	Financial distress	0.958*
(Hájek and Henriques, 2024)	FinBERT senti-ment	Acquisition occurrence	N/A (F1 0.71*)
This study - M1	**XGBoost (Fin.)**	**Binary CAR direction**	**0.5408**
This study - M3	**Multimodal Fu-sion**	**Binary CAR direction**	**0.5655**

The comparison in Table 12 requires careful interpretation. The AUC range reported by Zhang *et al.* (2024) (0.56-0.58) applies to proxies for long-term deal success, not short-term announcement CAR direction, making direct comparison difficult despite the overlapping numerical values. Similarly, high accuracy figures from Elhoseny *et al.* (2022) and high F1 figures from Hájek and Henriques (2024) address different, structurally easier prediction targets - financial distress detection and deal occurrence classification - rather than the genuinely harder problem of binary CAR direction. The ceiling this study pushes against is the approximately 0.55 AUC limit on ratio-based M&A models, not the performance of architectures optimised for different financial tasks. Against that correct benchmark, the M3 gain of +0.0247 is meaningful and directionally consistent with the theoretical prediction.

5.2.3 Product Limitations

The product should be evaluated against the standards of a deployable decision tool as well as an academic contribution. Against that second standard, three limitations are significant.

AUC and deployment readiness. An AUC of 0.5655, while better than the documented tabular ceiling for this exact prediction task, does not reach the performance threshold that an investment bank or advisory firm would require before incorporating a model into live deal screening. Commercial M&A analytics tools typically target AUC above 0.70 for directional predictions with real capital at stake. The product as built is a proof of concept establishing that multimodal signal exists and is structurally recoverable - it is not a production system. A dissertation-scale study produces a proof of concept establishing that multimodal signal exists; it is not designed to meet the deployment thresholds of commercial analytics platforms.

Text architecture incompleteness. The section-aware FinBERT pipeline demonstrates that section semantics matter and that undifferentiated text is harmful, but the actual implementation still compresses each section into PCA-reduced vectors before fusion. A richer implementation would apply cross-section attention, learning the interaction between MD&A and Risk Factor signals within a unified text encoder rather than treating them as independent scalar contributions. The current architecture makes the theoretical argument but does not fully execute the representational sophistication it implies.

Graph coverage sparsity. Bloomberg SPLC and related data sources provide reliable supply-chain relationships for large, publicly traded firms with extensive analyst coverage. Mid-cap and smaller acquirers frequently have partial or absent graph coverage. The product's performance on deals involving structurally well-documented acquirers may not generalise to the long tail of smaller transactions where graph data are sparse, creating a selection bias toward deals where the product works best. Future iterations must integrate alternative graph construction methods-such as extracting supplier-customer linkages from automated parsing of earnings call transcripts or proprietary Private Equity datasets-to resolve this sparsity.

5.2.4 The Deal Intelligence Terminal as Product

The Deal Intelligence Terminal deserves evaluation as a distinct product component. Its role is not merely to visualise results - it constitutes the operational proof that the multimodal architecture can be presented interactively to non-technical stakeholders without sacrificing rigour. The terminal's five phases (Data Profile, Model Evidence, SHAP Attribution, Hypothesis Lab, and Scenario Analysis) map directly onto the five empirical claims of the dissertation, ensuring that every assertion made in the text has a dynamic, interrogable visual counterpart.

The product succeeds here. Every hypothesis result, SHAP decomposition, and ablation comparison is rendered in real time from the underlying results JSON files, meaning that any examiner or future researcher who modifies the data pipeline will see updated results immediately without re-editing the text. This reproducibility property exceeds what is typically achieved in static academic reports and constitutes a secondary product contribution beyond the model itself.

5.3 Process Evaluation

5.3.1 Research Design Decisions That Worked

Several design decisions proved more valuable in practice than their theoretical justification alone suggested.

Binary classification over continuous regression. The decision to classify CAR direction rather than predict its magnitude was theoretically motivated in Chapter 2 but repeatedly validated in practice. Every time a regression model was tested on continuous CAR values, negative R^2 values confirmed that magnitude is dominated by noise. Had the study committed to regression as the primary objective, the project would have generated results too weak to build a coherent argument around. The binary formulation created a tractable problem precisely because it is harder to drown a directional signal in noise than a magnitude signal.

Late fusion architecture. The decision to encode each modality independently and fuse downstream rather than training end-to-end proved correct for the dataset available. With 2,864 complete multimodal observations, end-to-end training would almost certainly have collapsed into memorisation of training noise rather than genuine representation learning. Late fusion imposed a useful discipline: each modality had to be meaningful on its own before being combined, and the ablation ladder directly tests that property.

Treating the M2 reversal as a finding, not a failure. In a less carefully designed study, the M2 AUC drop would have been hidden, smoothed, or blamed on a FinBERT implementation bug. Elevating it to a first-class theoretical finding - that undifferentiated text actively destroys predictive value - transforms what could have been an embarrassment into one of the study's clearest and most defensible contributions. This required the intellectual courage to report a negative result prominently rather than burying it in an appendix.

Leakage control discipline. Fitting all preprocessing within the cross-validation loop rather than globally was operationally expensive but scientifically essential. Early experiments revealed that globally-fitted scalers inflated validation AUC non-trivially. Enforcing per-fold fitting reduced reported performance but produced estimates that are, to the extent possible given the data, realisable in deployment. A study that inflates performance through silent leakage produces findings that are formally unpublishable and practically misleading.

5.3.2 Research Design Decisions That Underperformed

An honest process evaluation also requires identifying where the process fell short of what was possible.

Hyperparameter search was counterproductive. The effort invested in Bayesian search via Optuna (Akiba *et al.* (2019)) yielded a critical domain-specific finding: surrogate-based optimizers systematically over-regularize in high-noise financial regimes. While it reduced AUC across every configuration, this provided a valuable methodological lesson: in noisy financial signal environments, architectural diversity (what features you include) is more valuable than optimiser sophistication (how finely you tune the same feature set). Future work should invest optimiser effort only after establishing that the feature space contains enough signal to reward fine-grained search.

Graph coverage could have been deeper. The HeteroGraphSAGE component achieves the headline AUC gain, but the graph neighbourhood depth was limited to two hops due to computational constraints. Economic theory suggests that second and third-order supplier dependencies - the suppliers of suppliers - carry meaningful risk propagation signals that two-hop neighbourhoods capture only partially. The gain demonstrated is therefore likely a lower bound on what a fully realised graph component could achieve with deeper neighbourhood sampling or a richer edge-weighting scheme.

PCA on FinBERT embeddings. This was evaluated as a defensive design choice in Chapter 3 and that assessment stands. But the process evaluation adds a practical observation: the PCA compression was decided at the architecture design stage primarily on theoretical grounds, and its empirical effect was never rigorously ablated independently of the other text components. A cleaner experimental design would have tested compressed versus uncompressed embeddings in isolation at a stage where their independent effect could be measured, rather than evaluating them only within the full text block. This gap in the ablation ladder means it is not possible to quantify how much predictive value the PCA step sacrificed. However, this represents a natural limitation of a full-scope study at dissertation scale, and constitutes a clear methodological agenda item for a journal extension.

5.3.3 Scope and Time Management

The project scope was ambitious relative to the time and data constraints of a BSc dissertation. Integrating three data environments (Yahoo Finance, Bloomberg SPLC, SEC EDGAR), building a full feature engineering pipeline across three modalities, training and evaluating multiple model families under proper cross-validation, implementing three formal hypothesis tests, building a production-quality interactive dashboard, and writing an academic dissertation simultaneously represents a substantial engineering and research undertaking.

The consequence of this scope is that depth in certain areas was necessarily sacrificed for breadth. The graph component could have been more deeply developed; the text pipeline could have included a richer dual-stream encoder; the regression analysis could have been more systematically reported. Each of these represents a genuine product limitation, and each originates in a process decision to maintain the full three-modality scope rather than narrowing to one or two modalities and pursuing them more thoroughly.

Whether this was the correct trade-off is debatable. The broadest intellectual contribution - that multimodal fusion combining financial, textual, and topological signals improves M&A prediction - required all three modalities to be present in the experiment. Removing any one of them would have prevented the ablation ladder from demonstrating the specific contribution of each stream. The scope was therefore not incidental; it was structurally necessary for the central argument. The cost was depth of execution in each individual stream.

5.3.4 Reflections on the Research Question

Looking back across the full project, the research question - whether heterogeneous graph topology adds predictive signal to M&A outcome classification beyond financial and textual baselines - was well-posed and proved to be answerable at dissertation scale. The answer is yes, and the evidence is sufficiently clear that it survives the most obvious methodological objections.

What was perhaps underestimated at the outset was how much of the project's intellectual value would come from negative results: the M2 reversal showing that naive NLP hurts, the negative R^2 values confirming that continuous CAR regression is intractable, and the hyperparameter search confirming that tuning without architectural improvement produces nothing. These negative findings are collectively as valuable as the positive AUC gain, because they precisely specify the boundary conditions under which multimodal fusion succeeds and fails. A project that produced only positive results without these boundary tests would have made a narrower and less credible contribution.

The study therefore succeeds not merely because AUC improved, but because it provides a structured empirical account of what types of information help, what types hurt, and why the architectural choices made in Chapter 3 were necessary rather than optional. Reporting the parts that didn't work, like the regression results and the M2 reversal, made the final AUC gain feel much more like a real finding rather than just a fluke.

5.4 Personal Reflection and Mental Wealth

Building a multimodal model for M&A wasn't just a technical challenge; it was an exercise in managing the frustration that comes with high-stakes data engineering. Over the course of the project, I had to completely change how I approached technical failure and personal productivity to meet the Mental Wealth requirements of the course.

The most difficult period was midway through the implementation, when I was struggling to merge the Bloomberg SPLC data with the SEC filing archives. I hit a series of "silent" bugs where deals were being dropped because of minor ticker mismatches, and for a while, it felt like the more data I added, the worse the model performed. The "Python 3.14 freeze" I encountered during the final training run was a breaking point—I spent twelve hours trying to fix a compilation error that turned out to be a known environment bug. Initially, I took these roadblocks personally, feeling that my inability to "just fix it" meant I wasn't up to the task.

However, the real breakthrough in my personal development came when I stopped trying to brute-force every problem. I had to unlearn the "hacker" habit of working until 3:00 AM on four cups of coffee. I started setting hard cut-off times for coding and forced myself to take breaks even when a bug wasn't fixed. This sounds simple, but for a project this size, it was the only way to stay sharp enough to spot the "M2 reversal" pattern in the results. I realized that the negative results—like the failure of the regression models—weren't personal failures; they were actually some of the most interesting findings in the whole study. Learning to step away from the terminal and look at the project with a rested mind was probably the most valuable skill I gained, and it's something I'll take with me into my professional career.

Chapter 6: Conclusion

6.1 The Core Argument Resolved

The enduring failure of quantitative M&A prediction is rooted in economic epistemology rather than computational limits — models have had enough power; they have lacked the right representation of the firm. For three decades, quantitative finance has treated the firm as an isolated vector of accounting ratios - a standalone entity disconnected from its ecosystem. This dissertation began with a single structural critique: that the tabular paradigm enforces an artificial independence assumption upon firms that are deeply, inextricably networked. The central thesis argued that until predictive models encode the economic reality of supply chains, competitive topologies, and strategic semantics, the accuracy ceiling on M&A prediction would remain mathematically unbroken.

This study resolved that argument by building and evaluating a multimodal late-fusion architecture (HeteroGraphSAGE) designed specifically to break the tabular ceiling. By fusing conventional financial data with section-aware textual embeddings and structural graph topology, the project proved that post-acquisition cumulative abnormal return (CAR) direction is predictable not just from what firms earn, but from where they sit in the industrial network and how they articulate their strategic vulnerabilities.

6.2 Synthesis of Empirical Findings

The empirical findings of this research fundamentally reshape how M&A prediction should be approached, distinguishing clearly between what adds predictive value and what destroys it.

The most prominent positive result was the validation of the Topological Alpha Hypothesis (H1). The addition of graph structural features elevated the classification AUC from the financial-only baseline of 0.5408 (M1) to 0.5655 (M3) - a non-trivial $+0.0247$ lift in a domain defined by severe market noise (Betton, Eckbo and Thorburn, 2008). This confirms that supply chain proximity and network positioning contain irreducible economic signals that tabular representations categorically miss (Cohen and Frazzini, 2008; Ahern and Harford, 2014). Furthermore, the Topological Arbitrage Hypothesis (H3) demonstrated that network centrality acts as a structural variance dampener ($p = 0.0079$), establishing that highly interconnected firms experience less volatile market reactions because their exposure is diversified and their integration logic is more legible to the market (Larcker, So and Wang, 2013).

Equally important were the boundary conditions established by the negative findings. The Semantic Divergence Hypothesis (H2) and the subsequent “M2 Reversal” proved that applying natural language processing naively - by aggregating corporate disclosures into a single semantic blob - actively destroys predictive accuracy. Because Management Discussion & Analysis (MD&A) and Risk Factor disclosures encode opposing economic forces, they must be modelled as distinct, adversarial signals. Additionally, the uniform failure of the regression pipeline to explain continuous CAR magnitude ($R^2 < 0$) confirmed that while the **direction** of synergy is probabilistically recoverable through structural data, the precise **magnitude** of announcement returns remains structurally dominated by idiosyncratic market noise.

6.3 Research Contributions

This dissertation makes three primary contributions to the intersection of data science and financial economics:

1. **Methodological Innovation:** It introduces one of the first explicit applications of heterogeneous graph neural networks (HeteroGraphSAGE) to the specific problem of post-merger synergy outcome classification, formally moving the field beyond the tabular independence assumption.
2. **Empirical Deconstruction of Financial NLP:** It provides empirical proof that section-level semantic divergence matters in M&A prediction, establishing a clear methodological warning against the deployment of unpartitioned textual embeddings in corporate finance.
3. **The Deal Intelligence Terminal:** It operationalises the theoretical framework into an interactive, fully reproducible research artefact, proving that complex multimodal predictions and SHAP-based interpretations can be rendered transparently for decision-makers.

6.4 Future Opportunities

The limitations identified in Chapter 5 dictate the immediate path forward for subsequent research. The architecture developed here is foundational but not exhaustive, leaving three distinct avenues for expansion.

First, the textual pipeline - currently reliant on late-fusion PCA compression - should be upgraded to a cross-attention transformer architecture. Allowing the model to dynamically attend to the interaction between the acquirer's strategy and the target's risk factors at the token level, rather than the document level, will likely recover the semantic signal currently lost to dimensionality reduction.

Second, the graph topology was constrained to static, two-hop supplier-competitor networks. Future architectures should temporalise the graph, allowing the network to evolve continuously leading up to the announcement date. Incorporating dynamic edge weights that reflect the volume and frequency of supply-chain transactions would transition the model from measuring mere connectivity to measuring actual economic flow.

Third, while this study utilised Cumulative Abnormal Return (CAR) as the sole market-based proxy for synergy, future models should incorporate a multi-horizon objective that simultaneously predicts short-window CAR and long-term accounting performance (e.g., Return on Invested Capital over a three-year horizon). This would explicitly test whether the topological structures that please the market at announcement actually deliver the operational efficiencies promised.

6.5 Reflection on Aims, Objectives, and Personal Development

6.5.1 Achievement of Objectives and Project Trajectory

The project successfully achieved its original aim of developing a multimodal framework for M&A outcome prediction, though the trajectory matured significantly from the initial proposal. The original objectives focused primarily on integrating financial data with textual sentiment. However, as the literature review progressed, it became evident that simply adding text was insufficient without preserving section-level semantics (the M2 reversal finding), and that supply-chain network topology was a critical missing dimension. Consequently, the objectives evolved to include heterogeneous graph neural networks (HeteroGraphSAGE), elevating the project from a standard NLP integration task to a tri-modal fusion architecture.

The evaluation objective was successfully met by establishing a strict ablation ladder, proving that topological data yields orthogonal predictive alpha. Furthermore, the interpretability objective was achieved using SHAP decomposition, providing economic credibility to the model's outputs.

6.5.2 Difficulties and Learning Curve

The learning curve for this project was exceptionally steep. The integration of three distinct data environments-Yahoo Finance, SEC EDGAR, and Bloomberg SPLC-posed significant data engineering challenges. Cleaning and aligning temporal datasets to enforce strict event-window embargoes required meticulous programmatic discipline to prevent look-ahead contamination. Architecturally, transitioning from standard tabular machine learning to implementing PyTorch-based HeteroGraphSAGE and fine-tuning FinBERT pipelines pushed my technical capabilities significantly. Dealing with negative results, such as the initial failure of continuous CAR regression models, was initially frustrating but ultimately became one of the study's strongest methodological findings, reinforcing the value of honest scientific reporting over artificial metric-chasing.

6.5.3 Real-World Use Case and Transferable Skills

Beyond the academic contribution, this dissertation has equipped me with highly transferable skills directly applicable to industry. The construction of the Deal Intelligence Terminal demonstrated my ability to operationalise complex machine learning models into interactive, production-ready full-stack applications. The rigorous approach to data leakage, cross-validation, and financial metric standardisation mirrors the strict requirements of quantitative finance and algorithmic trading desks. Furthermore, managing the scope of a large-scale data science project-balancing computational resource limits with architectural ambition-has refined my project management and problem-solving resilience. These capabilities bridge the gap between theoretical data science and deployable AI engineering, positioning me strongly for roles in ML architecture, quantitative analysis, and full-stack software development.

6.6 Final Remark

The prediction of corporate value creation can no longer operate under the fiction of the isolated firm. As global supply chains become more fragile and corporate ecosystems become more integrated, the models we use to evaluate their convergence must respect that complexity. The tabular ceiling was always a representational limit, not a computational one — and closing that gap required modelling firms as the networked economic actors they actually are. By demonstrating that graph topology and section-aware semantics carry irreducible predictive value, this study establishes that the future of financial machine learning lies not in squeezing more signal from the balance sheet, but in mapping the complex, interconnected reality of the firm itself.

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